变系数非线性波动方程（二阶拟线性双曲型 PDE）

①Kirchhoff 弹性杆理论(几何精确)

② Maxwell 磁场张量理论(电磁力)

③ 修正 Morison 方程(流圈耦合)

④ 接触力学

推导过程如下：

(1)几何与运动学描述(① Kirohhoff 假设)

$\vec{r}(s,t)$ ： 杆的中心线位置矢量

s:参考弧长坐标（Lagrange 材料坐标）.

t：时间

$\vec{\tau}$:变形后的单位切向量

λ：中心线伸长率（拉伸应变）

$$(\lambda(S,t):=\left\|\frac{\partial r(S,t)}{\partial S}\right\|)$$

$$(\boldsymbol{\tau}(S,t):=\frac{\partial_S r(S,t)}{\|\partial_S r(S,t)\|})$$

$$(\frac{\partial r}{\partial S}=\lambda\boldsymbol{\tau},\qquad \|\boldsymbol{\tau}\|=1)$$

(v)

$$\Rightarrow\frac{\partial\vec{r}}{\partial s}=v\vec{t},v=|\frac{\partial\vec{r}}{\partial s}|\approx 1$$

$v = 1 + \varepsilon$, $\vec{\mathrm{t}}$取成$\overrightarrow{\mathrm{ds}}$

$$\frac{\partial \mathbf{r}}{\partial S} = \lambda \boldsymbol{\tau}, \quad \lambda = \left|\frac{\partial \mathbf{r}}{\partial S}\right|, \quad |\boldsymbol{\tau}| = 1$$

$\lambda = 1 + \varepsilon$, $\vec{\tau}$取成$\overrightarrow{\mathrm{ds}}$

(vi)

$$\frac{\partial \vec{d}_i}{\partial s} = \vec{u} \times \vec{d}_i$$

$\vec{\mathrm{k}}$: 曲率矢量 $\approx \frac{\partial \vec{t}}{\partial s} \times \vec{t} = \frac{1}{v}\frac{\partial}{\partial s}\left(\frac{1}{v}\frac{\partial \vec{r}}{\partial s}\right) \times \frac{\partial \vec{r}}{\partial s}$

$\vec{u}$: Darboux 矢量轴, 描述杆的局部变形

$\{\vec{d}_1, \vec{d}_2, \vec{d}_3\}$: 局部标架

$$[\boldsymbol{\kappa} = \begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix}.]$$

$$[\hat{a}\, b = a \times b.]$$

$$[\boldsymbol{\kappa}^s = R\boldsymbol{\kappa}.]$$

$$[d_i \cdot d_j = \delta_{ij}, \qquad d_1 \times d_2 = d_3.]$$

$$[R = [d_1 \quad d_2 \quad d_3] \in SO(3).]$$

$$[R^T R = I, \qquad \det R = 1.]$$

$$\frac{\partial d_i}{\partial S} = \boldsymbol{\kappa}^s \times d_i, \qquad i = 1,2,3.$$

$$R_S = R\hat{\boldsymbol{\kappa}},$$

$$[\boldsymbol{\kappa}^s = R\boldsymbol{\kappa}.]$$

(2) **动力学平衡与本构关系**

(i)

$$\rho A \frac{\partial^2 \vec{r}}{\partial t^2} = \frac{\partial \vec{n}}{\partial s} + \vec{f}_{ext}$$

(牛顿第二定律. Euler-Cauchy 应力原理)

平动方程 $\vec{f}_{ext} = \vec{f}_{mag} + \vec{f}_{hydro} + \vec{f}_{cont} + \vec{f}_{gravity} = \vec{f}_{mag} + \vec{f}_{hydro} + \vec{f}_{cont}$

($\vec{f}_{gravity} = 0$：微重力)

$\vec{n}(s, t)$：内力矢量

$\vec{n}$:杆件横截面上的内力矢量

$\vec{m}$:杆件横截面上的力矩矢量

ρ, A, I: 杆的材料密度, 横截面积, 截面惯性矩

$\vec{f}_{mag} = \oint \vec{\sigma}_m \cdot \vec{n} dl$(Maxwell 磁场张量理论)磁力(4)(②证明公式含义)

$\vec{I}_m$:磁偶极矩

$\vec{\sigma}_m$:Maxwell 应力张量

(修正 Morison 方程计算得到的流体作用力)③

$$\vec{f}_{hydro} = -\rho_f A C_m (\frac{\partial^2 \vec{r}}{\partial t^2} - \vec{V}_f) - \frac{1}{2}\rho_f C_d |\vec{V}_r| \vec{V}_r \quad (5)$$

（惯性项+阻尼项）

$\vec{V}_f$:流体速度

ρ_f:流体密度

V:杆件体积

C_m:附加质量系数

C_d:拖曳系数

$\vec{V}_r$:杆件与流体的相对速度

$$\vec{f}_{cont} = \begin{cases} -k_p(d-d_0)^n \vec{n}_{wall} - \mu_f |\vec{F}_{normal}| \vec{t}_{\text{fric}}, & d \le d_0 \\ 0, & d > d_0 \end{cases} \quad (6)$$

（④接触力）

d:杆表面到障碍物的距离

d_0:接触判定阈值

k_p:接触刚度

n:指数(Hertz 模型常取 n=1.5)

$\vec{f}_{gravity}$(微重力环境下的残余重力)(7)

（ii）

$\frac{\partial \vec{m}}{\partial s} + \frac{\partial \vec{r}}{\partial s} \times \vec{n} + \vec{l}_{\text{mag}} + \vec{l}_{cont} = J \frac{\partial \vec{w}}{\partial t} + \vec{w} \times (J\vec{w})$ （转动方程）

$\vec{w} \times (J\vec{w})$:(非线性旋转效应)描述大变形时需要考虑这一项

$\vec{w}$: 杆件截面的角速度矢量

$$J = \rho I$$

g: 材料密度. kg/m^3

I：几何惯性致密， m^4

(iii)

力—应变关系

$$\vec{n} = K_s(\vec{r} - \vec{r_0})$$

$$\begin{cases} \vec{r}: \text{剪切/拉伸应变矢量} \\ K_s: \text{剪切/拉伸刚度矩阵} \end{cases}$$

力矩—曲率关系

(iv)

$$\vec{m} = K_b(\vec{u} - \vec{u_0})$$

$$\begin{cases} \vec{u} = (k_1, k_2, z)^T : \text{曲率与扭率矢量} \\ \mathrm{K_b} = \mathrm{diag}(EI_1, EI_2, EI_3,): \text{弯曲与扭转刚度矩阵} \end{cases}$$

将(i), (ii), (iii), (iv), (v), (vi)这 6 个方程整理成以下方程组：

Part1(四场耦合 PDE 建立)

$$\begin{cases} \rho A\frac{\partial^2\vec{r}}{\partial t^2} - \frac{\partial\vec{n}}{\partial s} = \vec{\mathrm{f}}_{\mathrm{mag}}(\mathrm{B}) + \vec{\mathrm{f}}_{\mathrm{hydro}}(\frac{\partial\vec{r}}{\partial t}, \frac{\partial^2\vec{r}}{\partial t^2}) + \vec{\mathrm{f}}_{\mathrm{cont}}(\vec{\mathrm{r}}) + \vec{\mathrm{f}}_{\mathrm{gravity}}(\vec{\mathrm{r}}) \\ J\frac{\partial\vec{w}}{\partial t} + \vec{w}\times(J\vec{w}) - \frac{\partial m}{\partial s} - \frac{\partial r}{\partial s}\times\vec{n} = \vec{I}_{mag}(\vec{B}) + \vec{I}_{cont}(\vec{r}) \end{cases} \quad (1)$$

$$\begin{cases} \vec{m} = K_b(\vec{u} - \vec{u_0}) \\ \vec{n} = K_s(\vec{r} - \vec{r_0}) \end{cases} \quad (2)$$

$$\begin{cases} \frac{\partial\vec{r}}{\partial s} = (1+\varepsilon)\vec{d_3} \\ \frac{\partial\vec{d_i}}{\partial s} = \vec{u}\times\vec{d_i} \end{cases} \quad (3)$$

强非线性,强拟合,混合-PDE 代数方程组.

四场耦合非线性偏微分方程组.

注：①-③页内容为大致推导过程需要清晰每个变量的含义,核对公式是否正确？

将内力矢量$\vec{n}$写成两部分

$$\vec{n} = EA(v-1)\vec{t} + \nabla s \cdot \vec{m} + \overrightarrow{K_s}r$$

r:小剪切假设,忽略

三个分别为拉伸力，弯曲力矩，剪切力

$$\approx EA(v-1)\vec{t} + \nabla s \cdot \vec{m}$$

$$= EA\left(|\frac{\partial\vec{r}}{\partial s}| - 1\right)\frac{1}{v}\frac{\partial\vec{r}}{\partial s} - \frac{\partial}{\partial s}(EI\frac{\partial^2\vec{r}}{\partial s^2})$$

$$= EA\left(|\frac{\partial\vec{r}}{\partial s}| - 1\right)\left|\frac{\partial\vec{r}}{\partial s}\right|^{-1}\frac{\partial\vec{r}}{\partial s} - \frac{\partial}{\partial s}(EI\frac{\partial^2\vec{r}}{\partial s^2}) := k_{\tan}(s,t,r)\frac{\partial\vec{r}}{\partial s} - \frac{\partial}{\partial s}(EI\frac{\partial^2\vec{r}}{\partial s^2})$$

$$\Rightarrow \frac{\partial\vec{n}}{\partial s} = \frac{\partial}{\partial s}[k_{\tan}(s,t,r)\frac{\partial\vec{r}}{\partial s}] - \frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\vec{r}}{\partial s^2}) \quad (8)$$

将(5),(8)代入方程(1)得

$$\rho A\frac{\partial^2\vec{r}}{\partial t^2} - \frac{\partial}{\partial s}[k_{\tan}(s,t,r)\frac{\partial\vec{r}}{\partial s}] + \frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\vec{r}}{\partial s^2}) + \rho_f AC_m\frac{\partial^2\vec{r}}{\partial t^2} - \rho_f AC_m\vec{V}_f$$
$$= \vec{\mathrm{f}}_{\mathrm{mag}}\ (\mathrm{B}) - \frac{1}{2}\rho_f C_d\left|\vec{\mathrm{V}}_r\right|\vec{V}_r + \vec{\mathrm{f}}_{\mathrm{cont}}(\vec{\mathrm{r}}) + \vec{\mathrm{f}}_{\mathrm{gravity}}(\vec{\mathrm{r}})$$

$$\Rightarrow (\rho A+\rho_f AC_m)\mathrm{I}\frac{\partial^2\vec{r}}{\partial t^2}-\rho_f AC_m$$
$$=\frac{\partial}{\partial s}[k_{\tan}(s,t,r)\frac{\partial\vec{r}}{\partial s}]-\frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\vec{r}}{\partial s^2})-\frac{1}{2}\rho_f DC_d\left|\frac{\partial\vec{r}}{\partial t}-\vec{V}_f\right|(\frac{\partial\vec{r}}{\partial t}-\vec{V}_f)$$
$$+\vec{\mathrm{f}}_{\mathrm{cont}}(\vec{\mathrm{r}})+\vec{\mathrm{f}}_{\mathrm{gravity}}(\vec{\mathrm{r}})+\vec{\mathrm{f}}_{\mathrm{mag}}\ (\mathrm{B})$$

$$\Rightarrow (\rho A+\rho_f AC_m\mathrm{I})\frac{\partial^2\vec{r}}{\partial t^2}+(\frac{1}{2}\rho_f DC_d\left|\frac{\partial\vec{r}}{\partial t}-\vec{V}_f\right|I)\frac{\partial\vec{r}}{\partial t}$$
$$=\frac{\partial}{\partial s}[k_{\tan}(s,t,r)\frac{\partial\gamma}{\partial s}]-\frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\vec{r}}{\partial s^2})+\rho_f AC_m\vec{V}_f+\vec{f}_{cont}(\vec{r})+\vec{f}_{gravity}(\vec{r})$$
$$+\vec{f}_{mag}\ (B)$$

$$\Rightarrow M(s,t)\frac{\partial^2\vec{r}}{\partial t^2}+C(s,t,\frac{\partial\vec{r}}{\partial t})\frac{\partial\vec{r}}{\partial t}=\frac{\partial}{\partial s}[k_{\tan}(s,t,r)\frac{\partial\gamma}{\partial s}]-\frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\vec{r}}{\partial s^2})+\vec{\mathrm{F}}(9)$$

$$M(s,t):=\rho A+\rho_f AC_m\mathrm{I}$$

$$C(s,t,\frac{\partial\vec{r}}{\partial t}):=\frac{1}{2}\rho_f DC_d\left|\frac{\partial\vec{r}}{\partial t}-\vec{V}_f\right|I$$

$$\vec{\mathrm{F}}:=\rho_f AC_m\vec{V}_f+\vec{f}_{cont}(\vec{r})+\vec{f}_{gravity}(\vec{r})+\vec{f}_{mag}\ (B)(9)$$

记为$L(s,t,\vec{r}(s,t))=\vec{\mathrm{F}}$

变系数非线性波动方程

近似解：

$\vec{r}(s,t)\approx N(s)\overrightarrow{q_k}(t):=\vec{r}(s,t),$

$\vec{r}(s,t)=\sum_{k=1}^{N}N_k(s)\overrightarrow{q_k}(t)$

s 与 t 分离，$s\in[0,\ L]$

$N(S)$ ： 已知的插值函数矩阵

$\overrightarrow{q_k}(t)$:未知的节点位移向量

残差$\vec{R}(s,t) = L(s,t,\vec{r}(s,t)) - \vec{\mathrm{F}} \neq 0$

Galerkin 空间,离散化,Part 2.

要求： $\int_0^L N^T(s)\cdot\vec{R}(s,t)ds \equiv 0$（残差在整体域上的加权积分为 0）

$$\Leftrightarrow \int_0^L N^T(s)\,[M\ddot{\hat{r}} + C\dot{\hat{\gamma}} - \frac{\partial}{\partial s}(k_{\tan}\frac{\partial\hat{\gamma}}{\partial s}) + \frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2}) - \vec{\mathrm{F}}]ds = 0 \quad (10)$$

$$\int_0^L N^T(s)\,M\ddot{\hat{r}}ds = \int_0^L N^T(s)\,MN(s)\ddot{\vec{q}_k}(t)ds = (\int_0^L N^T(s)\,MN(s)ds)\ddot{\vec{q}_k}(t)$$

$$\int_0^L N^T(s)\,C\dot{\hat{\gamma}}ds = \int_0^L N^T(s)\,CN(s)\dot{\vec{q}_k}(t)ds = (\int_0^L N^T(s)\,CN(s)ds)\dot{\vec{q}_k}(t)$$

$$\int_0^L N^T(s)\frac{\partial}{\partial s}(k_{\tan}\frac{\partial\hat{\gamma}}{\partial s})ds = N^T(s)k_{\tan}\frac{\partial\hat{\gamma}}{\partial s}\Big|_0^L - \int_0^L \frac{dN^T(s)}{ds}(k_{\tan}\frac{\partial\hat{\gamma}}{\partial s})ds$$

$$= N^T(s)k_{\tan}\frac{\partial\hat{\gamma}}{\partial s}\Big|_0^L - (\int_0^L \frac{dN^T(s)}{ds}k_{\tan}\frac{d\mathrm{N}(s)}{ds}ds)\vec{q_k}(t)$$

$$\int_0^L N^T(s)\frac{\partial^2}{\partial s^2}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2})ds = N^T(s)\frac{\partial}{\partial s}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2})\Big|_0^L - \int_0^L \frac{dN^T(s)}{ds}\frac{\partial}{\partial s}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2})ds$$

$$= N^T(s)\frac{\partial}{\partial s}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2})\Big|_0^L - \frac{dN^T(s)}{ds}EI\frac{\partial^2\hat{\gamma}}{\partial s^2}\Big|_0^L + \int_0^L \frac{d^2N^T(s)}{ds^2}EI\frac{\partial^2\hat{\gamma}}{\partial s^2}ds$$

$$= N^T(s)\frac{\partial}{\partial s}(EI\frac{\partial^2\hat{\gamma}}{\partial s^2})\Big|_0^L - \frac{dN^T(s)}{ds}EI\frac{\partial^2\hat{\gamma}}{\partial s^2}\Big|_0^L + (\int_0^L \frac{d^2N^T(s)}{ds^2}EI\frac{d^2\mathrm{N}(s)}{ds^2}ds)\vec{q_k}(t)$$

$$M_{global}: = \int_0^L N^T(s)\, MN(s) ds$$

$$C_{global}: = \int_0^L N^T(s)\, CN(s) ds$$

$$K_{ten}: = \int_0^L \frac{dN^T(s)}{ds} k_{\tan} \frac{d\mathrm{N}(s)}{ds} ds$$

$$K_{bent}: = \int_0^L \frac{d^2N^T(s)}{ds^2} EI \frac{d^2\mathrm{N}(s)}{ds^2} ds$$

$$\vec{F}_{global}: = \int_0^L N^T(s)\, \vec{F} ds$$

$$\Rightarrow M_{global}\ddot{\vec{q}_k}(t) + C_{global}\dot{\vec{q}_k}(t) \ + \ (K_{ten} + K_{bent})\vec{q_k}(t) \ = \vec{F}_{global}$$

简记成 $M\ddot{\mathrm{q}}(t) + C\dot{\mathrm{q}}(t) \ + \ K\mathrm{q}(t) \ = F$

$$M(t)\ddot{\mathrm{q}}(t) + C\big(t, \dot{\mathrm{q}}(t)\big)\dot{\mathrm{q}}(t) \ + \ K(t, \mathrm{q}(t))\mathrm{q}(t) \ = F(t)$$

几何非线性$N(s)$

$$M(t, q)\ddot{\mathrm{q}}(t) + C(t, q, \dot{q})\dot{\mathrm{q}}(t) \ + \ K(t, \mathrm{q})\mathrm{q}(t) \ = F(t, q)$$

大形变 $N(s,\, q)$

现在分析 ODEs:

$$M(t)\ddot{q}(t) + C(t,\dot{q}(t))\dot{q}(t) + K(t,q(t))q(t) = F(t)$$

更一般的形式为：

$$M(t,q(t))\ddot{q}(t) + C(t,q(t),\dot{q}(t))\dot{q}(t) + K(t,q(t))q(t) = F(t,q(t))$$

简记为：

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + K(q)q = F(q)$$

这几项依赖于什么，需要再次看之前的推导，特别是 $K(q)$ 这个矩阵

定义：

广义坐标(广义位移): $q \in \mathbb{R}^n$（节点位移矢量）

广义动量: $p \in \mathbb{R}^n$

$$p = \frac{\partial L}{\partial \dot{q}} = M(q)\dot{q} \quad \Rightarrow \quad \dot{q} = M^{-1}(q)p$$

$$H(q,p) = T(q,p) + V(q), \quad \text{（能量 = 动能 + 势能）}$$

$$T(q,p)=\frac{1}{2}\dot{q}^T M(q)\dot{q}=\frac{1}{2}(M^{-1}(q)p)^T M(q)\dot{q}=\frac{1}{2}p^T(M^{-1}(q))^T p=\frac{1}{2}p^T M^{-1}(q)p$$

(因为$M^{-1}(q)$对称）

(动能依赖于动量 p 和几何构型 q)

$M^{-1}(q)$为黎曼度规

$$V(q)=U_{stretch}(q)+U_{bend}(q)=\int K(q)\,q\,dq=\int(K_{tan}+K_{bend})q\,dq$$

$M^{-1}(q)$不是常矩阵，无法提到积分因子，所以不能写成$\frac{1}{2}q^T\mathrm{K}q$的形式

哈密顿正则方程

Q ：广义非保守力。

$$\begin{cases}\dot{q}=\dfrac{\partial H}{\partial p}\\ \dot{p}=-\dfrac{\partial H}{\partial q}+Q(q,\dot{q})\end{cases}$$

$$\Longrightarrow \dot{p}=-\frac{\partial(T(q,p)+V(q))}{\partial q}+Q(q,\dot{q})=-\frac{1}{2}p^T\left[\frac{\partial M^{-1}(q)}{\partial q}\right]p-\frac{dV(q)}{dq}+Q(q,\dot{q})$$

$$=-\frac{1}{2}p^T\left[\frac{\partial M^{-1}(q)}{\partial q}\right]p-K(q)q+Q(q,\dot{q})$$

$\frac{1}{2}p^T\left[\frac{\partial M^{-1}(q)}{\partial q}\right]$ p：几何惯性力（由于质量矩阵 M(q) 随位形 q 变化而产生）。

$K(q)q$：保守恢复力（源于弹性势能 V(q) 的梯度，K(q) 为刚度矩阵）。

$Q(q,\dot{q})$：非保守广义力。

$$Q(q,\dot{q})=F_{mag}+F_{cont}-c(q,\dot{q})\,\dot{q}=F_{mag}+F_{cont}-C(q,\dot{q})M^{-1}(q)p$$

$$\Rightarrow \begin{cases} \dot{q} = \dfrac{\partial H}{\partial p} \\ \dot{p} = -\dfrac{\partial H}{\partial q} + F_{mag} + F_{cont} - C\dfrac{\partial H}{\partial p} \end{cases}$$

ilton 结构的构造 part3

$$\Leftrightarrow \begin{cases} \dot{q} = \nabla_p H \\ \dot{p} = -\nabla_q H + F_{mag} + F_{cont} - C\nabla_p H \end{cases}$$

$$H(q,p) = \frac{1}{2}p^T M^{-1}(q)p + U_{stretch}(q) + U_{bend}(q)$$

$$\frac{dH}{dt} = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} = \frac{\partial H}{\partial q}\frac{\partial H}{\partial p} + \frac{\partial H}{\partial p}\left(-\frac{\partial H}{\partial q} + F_{ext} - C\frac{\partial H}{\partial p}\right) = \frac{\partial H}{\partial p}\left(F_{ext} - C\frac{\partial H}{\partial p}\right) \quad (12)$$

设计复值哈密顿神经网络 CHNN　Part 4

$$F_{\text{mag}} = (\vec{m} \cdot \nabla)\vec{B}(s,t)$$

$\vec{m}$：磁偶极矩

$\vec{B}$：磁场强度

∇：$\frac{\partial}{\partial x}\hat{\boldsymbol{i}} + \frac{\partial}{\partial y}\hat{\boldsymbol{j}} + \frac{\partial}{\partial z}\hat{\boldsymbol{k}}$磁场强度的梯度算子

推导：一个磁偶极矩为$\vec{m}$的物体，在外磁场 $\vec{B}$ 中，具有势能：$U = -\vec{m} \cdot \vec{B}$，在势场中受到的力等于势能的负梯度：（以下过程理解即可，无需写出来）

$$\vec{F} = -\nabla U = \vec{F} = -\nabla(-\vec{m} \cdot \vec{B}) = \nabla(\vec{m} \cdot \vec{B})$$

假设$\vec{m}$是常矢量，即 $\nabla\vec{m} = 0$。

利用矢量分析中的恒等式：

$$\nabla(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A})$$

已知$(\vec{B} \cdot \nabla)\vec{m} = 0, \vec{B} \times (\nabla \times \vec{m}) = 0$

$$\Longrightarrow \nabla(\vec{m} \cdot \vec{B}) = (\vec{m} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{m} + \vec{m} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{m}) = (\vec{m} \cdot \nabla)\vec{B} + \vec{m} \times (\nabla \times \vec{B})$$

$$\Rightarrow \vec{F} = \nabla(\vec{m} \cdot \vec{v})\vec{B} = (\vec{m} \cdot \nabla)\vec{B} + (\vec{m} \cdot \vec{v})\vec{B}$$

在静磁学条件（无电流或电流影响可忽略）下：$\nabla \times \vec{B} = 0$。

$$\Rightarrow \vec{F} = \nabla(\vec{m} \cdot \vec{B}) \approx (\vec{m} \cdot \nabla)\vec{B}$$

注：$\vec{m} \cdot \nabla = (m_x\vec{i} + m_y\vec{j} + m_z\vec{k}) \cdot \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}\right) = m_x\frac{\partial}{\partial x} + m_y\frac{\partial}{\partial y} + m_z\frac{\partial}{\partial z}$

$$\vec{F} = (\vec{m} \cdot \nabla)\vec{B} = \left(m_x\frac{\partial}{\partial x} + m_y\frac{\partial}{\partial y} + m_z\frac{\partial}{\partial z}\right)(B_x\vec{i} + B_y\vec{j} + B_z\vec{k})$$

$$\vec{F} = \left(m_x\frac{\partial B_x}{\partial x} + m_y\frac{\partial B_x}{\partial y} + m_z\frac{\partial B_x}{\partial z}\right)\vec{i} + \left(m_x\frac{\partial B_y}{\partial x} + m_y\frac{\partial B_y}{\partial y} + m_z\frac{\partial B_y}{\partial z}\right)\vec{j}$$
$$+ \left(m_x\frac{\partial B_z}{\partial x} + m_y\frac{\partial B_z}{\partial y} + m_z\frac{\partial B_z}{\partial z}\right)\vec{k}$$

$\vec{F}$ 源于磁场的不均匀性。若磁场$\vec{B}$不依赖于空间变量，则梯度$\vec{F} = 0$

$\vec{F}$的方向取决于 $\vec{m}$的方向和磁场 $\vec{B}$ 的空间变化率（梯度）

$\vec{m} \cdot \nabla$的物理意义：沿着 $\vec{m}$的方向去探测磁场的变化。

引入泊松括号

$f(q,p,t), g(q,p,t)$ 光滑函数

$q=(q_1 \cdots q_N), p=(p_1 \cdots p_N)$：广义坐标，广义动量

$$\{f,g\}=\sum_{i=1}^{N}\left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i}-\frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i}\right) \quad (14)$$

注：N：系统的自由度
q_i：第 i 个广义坐标
p_i：第 i 个广义动量。

$f(q,p,t)$是 t 的一元函数，也是 q, p, t 的三元函数。
全导数：

$$\frac{df}{dt}=\frac{\partial f}{\partial q}\frac{dq}{dt}+\frac{\partial f}{\partial p}\frac{dp}{dt}+\frac{\partial f}{\partial t}$$

若考虑 Hamiltonian 系统：

$$\begin{cases} dq=\dfrac{\partial H}{\partial p}dt \\ dp=-\dfrac{\partial H}{\partial q}dt \end{cases}$$

则

$$\frac{df}{dt}=\frac{\partial f}{\partial q}\cdot\frac{\partial H}{\partial p}-\frac{\partial f}{\partial p}\cdot\frac{\partial H}{\partial q}+\frac{\partial f}{\partial t}=\{f,H\}+\frac{\partial f}{\partial t}$$

$\{f,g\}$满足以下性质

1. 反对称性：$\{f,g\}=-\{g,f\}$
2. 线性性：$\{af+bg,h\}=a\{f,h\}+b\{g,h\}$
3. 莱布尼兹法则：$\{fg,h\}=f\{g,h\}+g\{f,h\}$
4. 雅可比恒等式：$\{f,\{g,h\}\}+\{g,\{h,f\}\}+\{h,\{f,g\}\}=0$

基本泊松括号的计算

$$\{q_i,q_j\}=\sum_{k=1}^{N}\left(\frac{\partial q_i}{\partial q_k}\frac{\partial q_j}{\partial p_k}-\frac{\partial q_i}{\partial p_k}\frac{\partial q_j}{\partial q_k}\right)=0$$

$$\{p_i,p_j\}=0$$

$$\{q_i,p_j\}=\sum_{k=1}^{N}\left(\frac{\partial q_i}{\partial q_k}\frac{\partial p_j}{\partial p_k}-\frac{\partial q_i}{\partial p_k}\frac{\partial p_j}{\partial q_k}\right)=\begin{cases}1, i=j\\0, i\neq j\end{cases}=\delta_{ij}$$

$$\{q_i,H\}=\sum_{k=1}^{N}\left(\frac{\partial q_i}{\partial q_k}\frac{\partial H}{\partial p_k}-\frac{\partial q_i}{\partial p_k}\frac{\partial H}{\partial q_k}\right)=\frac{\partial q_i}{\partial q_i}\frac{\partial H}{\partial p_i}-0=\frac{\partial H}{\partial p_i}$$

$$\{p_i,H\}=\sum_{k=1}^{N}\left(\frac{\partial p_i}{\partial q_k}\frac{\partial H}{\partial p_k}-\frac{\partial p_i}{\partial p_k}\frac{\partial H}{\partial q_k}\right)=0-\frac{\partial p_i}{\partial p_i}\frac{\partial H}{\partial q_i}=-\frac{\partial H}{\partial q_i}$$

$$\begin{cases}\{q_i,H\}:=\dfrac{\partial H}{\partial p_i}\\ \{p_i,H\}:=-\dfrac{\partial H}{\partial q_i}\end{cases}\Rightarrow\begin{cases}\dot{q}_i=\{q_i,H\}\\ \dot{p}_i=\{p_i,H\}\end{cases}\Rightarrow\begin{cases}\dot{q}=\{\dot{q},H\}\\ \dot{p}=\{\dot{p},H\}\end{cases}$$

令$\mathbf{z}=(\mathbf{q},\mathbf{p})^T$，则

$$\frac{dH}{dt}=\frac{\partial H}{\partial q}\dot{q}+\frac{\partial H}{\partial p}\dot{p}=\frac{\partial H}{\partial q}\frac{\partial H}{\partial p}+\frac{\partial H}{\partial p}\left(-\frac{\partial H}{\partial q}+F_{ext}-C\frac{\partial H}{\partial p}\right)=\frac{\partial H}{\partial p}\left(F_{ext}-C\frac{\partial H}{\partial p}\right)$$

$$\dot{\mathbf{z}}=\{\mathbf{z},H\}=\begin{pmatrix}0 & I\\ -I & 0\end{pmatrix}\begin{pmatrix}\dfrac{\partial H}{\partial p}\\ \dfrac{\partial H}{\partial q}\end{pmatrix}:=J\nabla H$$

显然

$$\nabla \mathrm{z}=\begin{pmatrix}\dfrac{\partial H}{\partial p}\\ \dfrac{\partial H}{\partial q}\end{pmatrix}=\mathrm{I}$$

方程(12)可进一步写成

$$\dot{\mathbf{z}}=\{\mathbf{z},H\}+\begin{pmatrix}0\\ F_{mag}+F_{cont}\end{pmatrix}-\begin{pmatrix}0\\ \mathrm{C}(\mathrm{q},\dot{\mathrm{q}})\nabla_{\mathbf{p}}H\end{pmatrix}$$

$$\Rightarrow\dot{\mathbf{z}}=\{\mathbf{z},H\}+\begin{pmatrix}0\\ F_{mag}+F_{cont}\end{pmatrix}+\begin{pmatrix}0 & 0\\ 0 & -\mathrm{C}(\mathrm{q},\dot{\mathrm{q}})\end{pmatrix}\begin{pmatrix}\nabla_{\mathbf{q}}H\\ \nabla_{\mathbf{p}}H\end{pmatrix}$$

$$:=\{\mathbf{z},H\}+\widetilde{F}_{ext}-D\nabla H=\{\mathbf{z},H\}+\begin{pmatrix}0\\ \nabla_{\mathbf{q}}\mathrm{G}_2\end{pmatrix}-D\nabla H=\{\mathbf{z},H\}+\begin{pmatrix}0 & 0\\ I & 0\end{pmatrix}\nabla\mathrm{G}_2-D\nabla H$$

引入耗散泊松括号

$$[f,H]=(\nabla f)^TD(\nabla H)$$

$\Rightarrow[\mathrm{z},H]=(\nabla\mathrm{z})^TD(\nabla H)=\mathrm{I}D(\nabla H)=D\nabla H$⑮

$\Rightarrow\dot{\mathbf{z}}=\{\mathbf{z},H\}-[\mathbf{z},H]+\widetilde{F}_{ext}=J\nabla H-D\nabla H+\widetilde{F}_{ext}=(J-D)\nabla H+\widetilde{F}_{ext}$:metriplectic dynamics

注： Coquinot, B. and Morrison, P.J. A general metriplectic framework with application to dissipative extended magnetohydrodynamics. J. Plasma Phys. 1-32, 2020.（可用结论，使用时再分析）

方程 (1.3) ：

$$\frac{\mathrm{d}f}{\mathrm{d}t}=[[f,\mathcal{F}]]=\{f,\mathcal{H}\}+\mathcal{T}(f,\mathcal{S})$$

f光滑函数

$\mathcal{H}$：哈密顿函数

$\mathcal{T}$：拉格朗日乘子

$\mathcal{S}$： $S=\int_\Omega\sigma$ 总的熵。它是驱动不可逆过程的势函数。它与泊松括号满足对称的退化条件，

是泊松括号的卡西米尔不变量（Casimir of the Poisson bracket）

$$\begin{cases} \{f,S\}=0, \quad \forall f \\ (H,S)=0, \quad \forall f \text{ , 退化性要求} \end{cases}$$

$$\begin{cases} \dfrac{dH}{dt}=\{H,H\}+T(H,S)=0+0=0 \\ \dfrac{dS}{dt}=\{S,H\}+T(S,S)=0+T(S,S)\geq 0 \end{cases}$$

构造复值生成函数（哈密顿量）：

$\mathcal{H}(q,p) = H(q,p) + i(G_1(q,p) + G_2(q,p)) := H(q,p) + iG(q,p)$(16)

$H(q,p)$是经典的实哈密顿量（能量）。

$G(q,p)$ 是一个实函数，与系统的耗散有关。

耗散力被纳入虚部G中

$$-C\nabla_p \mathrm{H} = \dot{p}_{diss},\ G_1(q,p) = \frac{1}{2}p^T C(q,\dot{q})p \Rightarrow \dot{p}_{diss} = -\nabla_p G_1$$

$$F_{mag}(\mathrm{q}) + F_{cont}(\mathrm{q}) = \dot{p}_{ext}$$

$F_{cont}(\mathrm{q})$可忽略

$$G_2(q) = \int F_{mag}\,(q)\,dq = (\vec{m}\cdot\nabla)\vec{B}(s,t)\,dq \Rightarrow \dot{p}_{ext} = -\nabla_q G_2$$

引入复坐标变换，将实相空间$((q,p) \in \mathbb{R}^{2N})$ 映射到复空间：这是一个正则变换，保持辛结构。相应的，偏导数算符变为

定义复坐标$\begin{cases}\xi = \frac{1}{\sqrt{2}}(q + ip) \\ \bar{\xi} = \frac{1}{\sqrt{2}}(q - ip)\end{cases}$(17)

Wirtinger 导数：$\begin{cases}\frac{\partial}{\partial \xi} = \frac{1}{2}\left(\frac{\partial}{\partial q} - i\frac{\partial}{\partial p}\right) \\ \frac{\partial}{\partial \bar{\xi}} = \frac{1}{2}\left(\frac{\partial}{\partial q} + i\frac{\partial}{\partial p}\right)\end{cases}$(18)

$(q,p) \to (\xi,\bar{\xi})$正则变换，保持辛结构

在$\mathbb{R}^N \times \mathbb{R}^N$ 实的相空间中，标准的辛形式为 $\omega = dq \wedge dp$， $\{q,p\} = 1$。

由(17)有$\begin{cases}q = \frac{1}{\sqrt{2}}(\xi + \bar{\xi}) \\ p = \frac{-i}{\sqrt{2}}(\xi - \bar{\xi})\end{cases} \Rightarrow \begin{cases}dq = \frac{1}{\sqrt{2}}(d\xi + d\bar{\xi}) \\ dp = \frac{-i}{\sqrt{2}}(d\xi - d\bar{\xi})\end{cases}$

$$\Longrightarrow \omega = dq \wedge dp = \left[\frac{1}{\sqrt{2}}(d\xi + d\bar{\xi})\right] \wedge \left[\frac{-i}{\sqrt{2}}(d\xi - d\bar{\xi})\right] = -\frac{i}{2}(d\xi + d\bar{\xi}) \wedge (d\xi - d\bar{\xi})$$

$$= -\frac{i}{2}(d\xi \wedge d\xi - d\xi \wedge d\bar{\xi} + d\bar{\xi} \wedge d\xi - d\bar{\xi} \wedge d\bar{\xi}) = -\frac{i}{2}(-d\xi \wedge d\bar{\xi} - d\xi \wedge d\bar{\xi})$$

$$= id\xi \wedge d\bar{\xi}$$

$id\xi \wedge d\bar{\xi}$是一个实值 2-形式

$$\{\xi,\bar{\xi}\} = \frac{\partial \xi}{\partial q}\frac{\partial \bar{\xi}}{\partial p} - \frac{\partial \xi}{\partial p}\frac{\partial \bar{\xi}}{\partial q} = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{-i}{\sqrt{2}}\right) - \left(\frac{i}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) = -\frac{i}{2} - \frac{i}{2} = -i$$

$$\frac{\partial \mathcal{H}}{\partial \bar{\xi}} = \frac{1}{2}\left(\frac{\partial}{\partial q} + i\frac{\partial}{\partial p}\right)(H + iG)$$
$$= \frac{1}{2}\left(\frac{\partial H}{\partial q} + i\frac{\partial G}{\partial q} + i\frac{\partial H}{\partial p} - \frac{\partial G}{\partial p}\right)$$
$$= \frac{1}{2}\left[\left(\frac{\partial H}{\partial q} - \frac{\partial G}{\partial p}\right) + i\left(\frac{\partial G}{\partial q} + \frac{\partial H}{\partial p}\right)\right]$$

$$\dot{\xi} = \frac{1}{\sqrt{2}}(\dot{q} + i\dot{p}) = \frac{1}{\sqrt{2}}\left[\frac{\partial H}{\partial p} + i\left(-\frac{\partial H}{\partial q} - c\frac{\partial H}{\partial p} + F_{ext}\right)\right] := \frac{1}{\sqrt{2}}\left[\left(\frac{\partial H}{\partial p} - i\frac{\partial H}{\partial q}\right) - ic\frac{\partial H}{\partial p} + F_{ext}\right] = \frac{1}{\sqrt{2}}\left[\left(\frac{\partial H}{\partial p} - i\frac{\partial H}{\partial q}\right) - i\nabla_p G_1 + \nabla_q G_2\right] \quad (19)$$

$$\frac{\partial H}{\partial \bar{\xi}} = \frac{1}{2}\left(\frac{\partial}{\partial q} + i\frac{\partial}{\partial p}\right)H = \frac{1}{2}\left(\frac{\partial H}{\partial q} + i\frac{\partial H}{\partial p}\right)H \Rightarrow -\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}} = \frac{1}{\sqrt{2}}\left(\frac{\partial H}{\partial p} - i\frac{\partial H}{\partial q}\right) \quad (20)$$

由前文（公式(18)）定义的 Wirtinger 导数关系出发：

$$\begin{cases} \dfrac{\partial}{\partial q}=\dfrac{\partial}{\partial \xi}+\dfrac{\partial}{\partial \bar{\xi}} \\ \dfrac{\partial}{\partial p}=\dfrac{1}{i}\left(\dfrac{\partial}{\partial \bar{\xi}}-\dfrac{\partial}{\partial \xi}\right)=i\left(\dfrac{\partial}{\partial \xi}-\dfrac{\partial}{\partial \bar{\xi}}\right) \end{cases}$$

代入(19)得

$$\dot{\xi}=\frac{1}{\sqrt{2}}\left[\left(\frac{\partial H}{\partial p}-i\frac{\partial H}{\partial q}\right)+\left(\frac{\partial}{\partial \xi}-\frac{\partial}{\partial \bar{\xi}}\right)G_1+(\frac{\partial}{\partial \xi}+\frac{\partial}{\partial \bar{\xi}})G_2\right]$$

用(20)的条件，有

$$=-\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}}+\frac{1}{\sqrt{2}}\left(\frac{\partial G_1}{\partial \xi}-\frac{\partial G_1}{\partial \bar{\xi}}\right)+\frac{1}{\sqrt{2}}(\frac{\partial G_2}{\partial \xi}+\frac{\partial G_2}{\partial \bar{\xi}})$$

推导结果（图中红色框内公式修正后）：

$$\dot{\xi}=-\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}}+\frac{1}{\sqrt{2}}\left(\frac{\partial(G_1+G_2)}{\partial \xi}-\frac{\partial(G_1-G_2)}{\partial \bar{\xi}}\right)=-\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}}+\frac{1}{2}\mathrm{P}(\xi,\bar{\xi}) \quad (21)$$

$$(\frac{1}{\sqrt{2}}\left(\frac{\partial(G_1+G_2)}{\partial \xi}-\frac{\partial(G_1-G_2)}{\partial \bar{\xi}}\right)\triangleq \mathrm{P}(\xi,\bar{\xi}))$$

耗散哈密顿系统

$$\begin{aligned}\frac{dH}{dt}&=\frac{\partial H}{\partial \xi}\dot{\xi}+\frac{\partial H}{\partial \bar{\xi}}\dot{\bar{\xi}}\\ &=\frac{\partial H}{\partial \xi}[-\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}}+\frac{1}{\sqrt{2}}\left(\frac{\partial(G_1+G_2)}{\partial \xi}-\frac{\partial(G_1-G_2)}{\partial \bar{\xi}}\right)]+\frac{\partial H}{\partial \bar{\xi}}[\sqrt{2}i\frac{\partial H}{\partial \xi}\\ &+\frac{1}{\sqrt{2}}\left(\frac{\partial(G_1+G_2)}{\partial \bar{\xi}}-\frac{\partial(G_1-G_2)}{\partial \xi}\right)]\\ &=\frac{1}{\sqrt{2}}[\frac{\partial H}{\partial \xi}\left(\frac{\partial(G_1+G_2)}{\partial \xi}-\frac{\partial(G_1-G_2)}{\partial \bar{\xi}}\right)+\frac{\partial H}{\partial \bar{\xi}}\left(\frac{\partial(G_1+G_2)}{\partial \bar{\xi}}-\frac{\partial(G_1-G_2)}{\partial \xi}\right)]\\ &=\frac{2}{\sqrt{2}}\mathrm{Re}[\frac{\partial H}{\partial \xi}\left(\frac{\partial(G_1+G_2)}{\partial \xi}-\frac{\partial(G_1-G_2)}{\partial \bar{\xi}}\right)]\end{aligned}$$

$$:=\frac{2}{\sqrt{2}}\mathrm{Re}[\frac{\partial H}{\partial \xi}\mathrm{P}(\xi,\bar{\xi})]\begin{cases}\leq 0, \text{耗散}\\ =0, \text{增益}\\ \geq 0, \text{保守哈密顿系统}\end{cases}$$

进行正则变换 $(\xi,\bar{\xi})\rightarrow(I,\theta)$

$$\begin{cases}\xi=\sqrt{I}\,e^{i\theta}\\ \bar{\xi}=\sqrt{I}\,e^{-i\theta}\end{cases} \quad (22)$$

验证辛形式不变

$$d\xi \wedge \mathrm{d}\bar{\xi} = \left(\frac{dI}{2\sqrt{I}}e^{i\theta} + \sqrt{I}\,ie^{i\theta}d\theta\right) \wedge \left(\frac{dI}{2\sqrt{I}}e^{-i\theta} + \sqrt{I}\,(-i)e^{-i\theta}d\theta\right)$$

$$= \frac{dI}{2\sqrt{I}}e^{i\theta} \wedge \frac{dI}{2\sqrt{I}}e^{-i\theta} + \frac{dI}{2\sqrt{I}}e^{i\theta} \wedge \sqrt{I}\,(-i)e^{-i\theta}d\theta + \sqrt{I}\,ie^{i\theta}d\theta \wedge \frac{dI}{2\sqrt{I}}e^{-i\theta}$$

$$+ \sqrt{I}\,ie^{i\theta}d\theta \wedge \sqrt{I}\,(-i)e^{-i\theta}d\theta)$$

$$= \frac{dI}{2\sqrt{I}}e^{i\theta} \wedge \sqrt{I}\,(-i)e^{-i\theta}d\theta + \sqrt{I}\,ie^{i\theta}d\theta \wedge \frac{dI}{2\sqrt{I}}e^{-i\theta} = \frac{-i}{2}dI \wedge d\theta + \frac{i}{2}d\theta \wedge dI$$

$$= -\,idI \wedge d\theta$$

$$\Rightarrow \omega = d\mathrm{q} \wedge \mathrm{d}p = d\xi \wedge \mathrm{d}\bar{\xi} = dI \wedge d\theta$$

变换是正则的。

由 22 有

$$\begin{cases} I = |\xi|^2 = \xi\bar{\xi} \\ \theta = \dfrac{1}{2i}\ln\dfrac{\xi}{\bar{\xi}} \end{cases}$$

$$\frac{\partial}{\partial\xi} = \frac{\partial}{\partial I}\frac{\partial I}{\partial\xi} + \frac{\partial}{\partial\theta}\frac{\partial\theta}{\partial\xi} = \bar{\xi}\frac{\partial}{\partial I} + \frac{1}{2i}\frac{\bar{\xi}}{\xi}\frac{1}{\bar{\xi}}\frac{\partial}{\partial\theta} = \sqrt{I}\,e^{-i\theta}\frac{\partial}{\partial I} + \frac{1}{2i\sqrt{I}\,e^{i\theta}}\frac{\partial}{\partial\theta} = \frac{e^{-i\theta}}{2\sqrt{I}}(2I\,\frac{\partial}{\partial I} - i\frac{\partial}{\partial\theta})$$

$$\frac{\partial}{\partial\bar{\xi}} = \frac{e^{i\theta}}{2\sqrt{I}}(2I\,\frac{\partial}{\partial I} + i\frac{\partial}{\partial\theta})$$

$$\Rightarrow \mathrm{P}(\xi,\bar{\xi}) = \sqrt{2}[\frac{\partial(G_1+G_2)}{\partial\xi} - \frac{\partial(G_1-G_2)}{\partial\bar{\xi}}]$$

$$= \sqrt{2}\{\frac{e^{-i\theta}}{2\sqrt{I}}(2I\,\frac{\partial(G_1+G_2)}{\partial I} - i\frac{\partial(G_1+G_2)}{\partial\theta}) - \frac{e^{i\theta}}{2\sqrt{I}}(2I\,\frac{\partial(G_1-G_2)}{\partial I} + i\frac{\partial(G_1-G_2)}{\partial\theta})\} := \widehat{P}(I,\theta)$$

$$\dot{I} = (\xi\dot{\bar{\xi}}) = \dot{\xi}\bar{\xi} + \xi\dot{\bar{\xi}} = (-\sqrt{2}\,i\frac{\partial H}{\partial\bar{\xi}} + \frac{1}{2}\mathrm{P}(\xi,\bar{\xi}))\bar{\xi} + \xi(\sqrt{2}\,i\frac{\partial H}{\partial\xi} + \frac{1}{2}\overline{\mathrm{P}(\xi,\bar{\xi})})$$

$$= (-\sqrt{2}\,i\frac{\partial H}{\partial I}\xi + \frac{1}{2}\mathrm{P}(\xi,\bar{\xi}))\bar{\xi} + \xi(\sqrt{2}\,i\frac{\partial H}{\partial I}\bar{\xi} + \frac{1}{2}\overline{\mathrm{P}(\xi,\bar{\xi})})$$

$$= \frac{1}{2}\mathrm{P}(\xi,\bar{\xi})\bar{\xi} + \frac{1}{2}\overline{\mathrm{P}(\xi,\bar{\xi})}\xi Re(\mathrm{P}(\xi,\bar{\xi})\bar{\xi}) = Re\big(\mathrm{P}(\xi,\bar{\xi})\sqrt{I}e^{-i\theta}\big)$$

$$= \sqrt{I}Re\big(e^{-i\theta}\mathrm{P}(\xi,\bar{\xi})\big) := \sqrt{I}Re\big(e^{-i\theta}\widehat{P}(I,\theta)\big)$$

$$e^{-i\theta}\mathrm{P}(\xi,\bar{\xi}) = \sqrt{2}\{\frac{e^{-2i\theta}}{2\sqrt{I}}(2I\,\frac{\partial(G_1+G_2)}{\partial I} - i\frac{\partial(G_1+G_2)}{\partial\theta}) - \frac{1}{2\sqrt{I}}(2I\,\frac{\partial(G_1-G_2)}{\partial I} + i\frac{\partial(G_1-G_2)}{\partial\theta})\}$$

$$\Rightarrow Re\big(e^{-i\theta}\mathrm{P}(\xi,\bar{\xi})\big)$$

$$= \sqrt{2}\{\cos(2\theta)\frac{2I}{2\sqrt{I}}\frac{\partial(G_1+G_2)}{\partial I} - \sin(2\theta)\frac{1}{2\sqrt{I}}\frac{\partial(G_1+G_2)}{\partial\theta} - \frac{2I}{2\sqrt{I}}\frac{\partial(G_1-G_2)}{\partial I}\}$$

$$\Rightarrow c = 2\sqrt{2}I\cos(2\theta)\frac{\partial(G_1+G_2)}{\partial I} - \sqrt{2}\sin(2\theta)\frac{\partial(G_1+G_2)}{\partial\theta} - 2\sqrt{2}I\frac{\partial(G_1-G_2)}{\partial I} := \varepsilon\mathrm{f}(I,\theta)$$

未检查

$$\theta = ?$$

$$\xi = \sqrt{I}e^{i\theta} \Rightarrow \dot{\xi} = \frac{\dot{I}}{2\sqrt{I}}e^{i\theta} + i\sqrt{I}\dot{\theta}e^{i\theta} \overset{二十一}{=} -\sqrt{2}i\frac{\partial H}{\partial I}\frac{\partial I}{\partial\bar{\xi}} + \frac{1}{2}\mathrm{P}(\xi,\bar{\xi})$$

$$=-\sqrt{2}iH'(I)\sqrt{I}e^{i\theta}+\frac{1}{2}\widehat{P}(I,\theta)\Rightarrow \text{左页}.\ \frac{\dot{I}}{2\sqrt{I}}+i\sqrt{2}\dot{\theta}=-\sqrt{2}iH'(I)\sqrt{I}+\frac{1}{2}e^{-i\theta}\widehat{P}(I,\theta)$$

$$\frac{\dot{I}}{2\sqrt{I}}+i\sqrt{I}\dot{\theta}=-\sqrt{2}i\dot{H}(I)\sqrt{I}+\frac{1}{2}\hat{p}(I,\theta)e^{-i\theta}\Rightarrow$$

$$\text{实部:}\begin{cases}\dfrac{\dot{I}}{\sqrt{I}}=R_e\left(\dfrac{1}{2}e^{-i\theta}\hat{p}(I,\theta)\right)\\ \sqrt{I}\dot{\theta}=-\sqrt{2}\dot{H}(I)\sqrt{I}+I_m\left(\dfrac{1}{2}e^{-i\theta}\hat{p}(I,\theta)\right)\end{cases}$$

$$\Rightarrow\begin{cases}\dot{I}=\sqrt{I}\,R_e\left(e^{-i\theta}\hat{p}(I,\theta)\right):=\varepsilon f(I,\theta)\\ \dot{\theta}=-\sqrt{2}\dot{H}(I)+\dfrac{1}{2\sqrt{I}}I_m\left(e^{-i\theta}\hat{p}(I,\theta)\right):=\omega(I)+\varepsilon g(I,\theta)\end{cases}$$

Part 4 提取近可积结构与本征频率

其中ε为小参数，表示扰动强度；$(\omega(I)=-\sqrt{2}H(I))$为未扰动系统的本征频率；$f,g$为扰动项。

2. 可积条件与非退化条件

注：当$\varepsilon=0$时，系统退化为可积系统：

$$\begin{cases}\dot{I}=0\\ \dot{\theta}=\omega(I)=-\sqrt{2}H'(I)\end{cases}$$

要求频率映射是非退化的，即：

$$\omega'(I)=-\sqrt{2}H''(I)\neq 0$$

Part 5 验证 KAM 定理三大条件

$$\forall|\langle k,\omega(I)\rangle|\geq\frac{\gamma}{|k|^{\tau}},$$

$$k\in\mathbb{Z}^N\{0\},\gamma>0,\tau>N-1,$$

$$|k|=|k_1|+\cdots+|k_N|$$

注：A Lecture on the Classical KAM Theorem" —— J. Pöschel, 2009. Page 6, Theorem.
必须要求扰动是实解析的（但如果哈密顿量 H 和扰动 P 是 C^r 光滑的，且光滑度 r 足够大，定理也可以适用。）

1. Moser, J. (1962). On invariant curves of area − preserving mappings of an annulus.
2. Pöschel, J. (1982). Invariant tori in Hamiltonian systems.
3. Kuksin, S. B. (1998). Nearly integrable infinite − dimensional Hamiltonian systems.
4. Betti, M. & Belle, P. (2004). KAM tori for dissipative PDEs.

(结论可以直接拿来使用)

回到设计 CHNN 主线 Part 4

Complex−Valued HNN: 复值哈密顿神经网络

用复变量统一描述机器人状态（位置 q，动量 p，相位α）.通过学习复值哈密顿量，实现能量守恒的精准动力学建模与控制，适用于涉及波动、相位或复杂耦合的机器人系统。

回顾

方程(15)：

$$\dot{z} = \{z, H\} - [z, H] + \widehat{F}_{ext} = J\nabla H - D\nabla H + \widehat{F}_{ext}$$

其中$D = D(z)$是耗散度量矩阵（对称正定张量）。

$\frac{dH}{dt} = (\nabla H)^T \dot{z} = (\nabla H)^T (J\nabla H - D\nabla H + \widehat{F}_{ext}) = (\nabla H)^T J\nabla H - (\nabla H)^T D\nabla H + (\nabla H)^T \widehat{F}_{ext}$(26)

$J^T = -J$反对称，$(\nabla H)^T J\nabla H = 0$

$$\forall x \neq 0, ((\mathrm{x})^T J\mathrm{x})^T = \mathrm{x}^T J^T \mathrm{x} = \mathrm{x}^T(-J)\mathrm{x} = -\mathrm{x}^T J\mathrm{x} \Rightarrow \mathrm{x}^T J\mathrm{x} = 0, \forall x \neq 0$$

不考虑外力$\widehat{F}_{ext}$时，若$(D(z))$正定，则$\frac{dH}{dt} = -(\nabla H)^T D\nabla H < 0$，系统能量衰减（耗散）

方程(21)：

$$\dot{\xi} = -\sqrt{2}i\frac{\partial H}{\partial \bar{\xi}} + \frac{1}{\sqrt{2}}\left(\frac{\partial(G_1 + G_2)}{\partial \xi} - \frac{\partial(G_1 - G_2)}{\partial \bar{\xi}}\right)$$

定义泊松括号：

为统一形式，定义复数空间中的泊松括号：

$$\{f, g\}_{\mathbb{C}} = -i\left(\frac{\partial f}{\partial \xi}\frac{\partial g}{\partial \bar{\xi}} - \frac{\partial f}{\partial \bar{\xi}}\frac{\partial g}{\partial \xi}\right)$$

据此，有$\{\xi, H\}_{\mathbb{C}} = -i\frac{\partial H}{\partial \bar{\xi}}$。?

(21)$\Rightarrow \dot{\xi} = \sqrt{2}\{\xi, H\}_{\mathbb{C}} + R(\xi, \bar{\xi})$(系数可调整)(24)

参考： “KAM Theory Meets Statistical Learning Theory: Hamiltonian Neural Networks with Non-zero Training Loss” — Y. Chen, T. Matsubara, T. Yaguchi. The 36th AAAI Conference on Artificial Intelligence (AAAI-22).

引入参数λ，用一个神经网络$\mathrm{H}_{\lambda}(\xi, \bar{\xi})$来拟合或近似真实的复值哈密顿量函数。

损失函数：需要设计并控制多个损失函数来训练网络：
1.能量损失 (Lenorgy)：保证能量演化符合物理规律。
2.雅可比损失 (Lacobi)：保证辛结构或某种守恒律。。
3.辛结构损失 (Lsymplectic)：保证学习到的动力学具有辛几何结构。

回顾方程(17):

$(q,p)\to(\xi,\bar{\xi})$正则变换，保持辛结构

$$\begin{cases}\xi=\dfrac{1}{\sqrt{2}}(q+ip)\\ \bar{\xi}=\dfrac{1}{\sqrt{2}}(q-ip)\end{cases}$$

回顾方程(21):

$$\dot{\xi}:=-\sqrt{2}i\frac{\partial H}{\partial\bar{\xi}}+\frac{1}{2}\mathrm{P}(\xi,\bar{\xi}):=-\sqrt{2}i(\frac{\partial H}{\partial\bar{\xi}}+\frac{i}{2\sqrt{2}}\mathrm{P}(\xi,\bar{\xi})):=-\sqrt{2}i(\frac{\partial H}{\partial\bar{\xi}}+i\frac{\partial\tilde{\mathrm{P}}(\xi,\bar{\xi})}{\partial\bar{\xi}})$$

$\dot{\xi}:=-\sqrt{2}i\frac{\partial(H+i\tilde{\mathrm{P}})}{\partial\bar{\xi}}:=-i\frac{\partial\tilde{\mathrm{H}}}{\partial\bar{\xi}}$(25)

$$\begin{cases}\tilde{\mathrm{H}}=\sqrt{2}(H+i\tilde{\mathrm{P}})\\ \tilde{\mathrm{H}}=\tilde{\mathrm{H}}(\xi,\bar{\xi})\end{cases}$$

H:保守.$i\tilde{\mathrm{P}}$:耗散+驱动,值小

其中：

$$\frac{i}{2\sqrt{2}}\mathrm{P}(\xi,\bar{\xi})=i\frac{\partial\tilde{\mathrm{P}}(\xi,\bar{\xi})}{\partial\bar{\xi}}\Leftrightarrow\frac{\partial\tilde{\mathrm{P}}(\xi,\bar{\xi})}{\partial\bar{\xi}}=\frac{1}{2\sqrt{2}}\mathrm{P}(\xi,\bar{\xi})\Leftrightarrow\tilde{\mathrm{P}}(\xi,\bar{\xi})=\frac{1}{2\sqrt{2}}\int\mathrm{P}(\xi,\bar{\xi})\,\mathrm{d}\bar{\xi}$$

$\tilde{\mathrm{P}}(\xi,\bar{\xi})$复变量且实值

从现在起考虑复值生成函数$\tilde{\mathrm{H}}(\xi,\bar{\xi})=\sqrt{2}(H(\xi,\bar{\xi})+i\tilde{\mathrm{P}}(\xi,\bar{\xi}))$

（注:与方程 16 雷同，但经过转化得到的形式）

根据您提供的图片，已提取文字公式内容如下。对手写字符进行了必要的辨识和修正，并对可能的歧义处进行了说明。

引入参数 λ，考虑拟合 $Hamiltonian(H_\lambda(z,\bar{z}))$。

引入参数 λ，考虑拟合 Hamiltonian $\widetilde{H_\lambda}(\xi,\bar{\xi})$

(i)损失函数

(**a**) 能量损失函数：

$$L_{energy}\overset{26}{\Rightarrow}\left\|\frac{d\widetilde{H}_\lambda(\xi,\bar{\xi})}{dt}\right\|^2=\left\|\sqrt{2}\frac{d\left[H_\lambda(\xi,\bar{\xi})+i\widetilde{P}_\lambda(\xi,\bar{\xi})\right]}{dt}\right\|^2$$

$$=2\left\|\frac{d\widetilde{P}_\lambda(\xi,\bar{\xi})}{dt}\right\|^2\quad(\text{能量}-\text{耗散行为约束})$$

(**b**)哈密顿损失函数：

$$L_{hamilton}=\left\|\dot{\xi}+i\frac{\partial\widetilde{H}_\lambda(\xi,\bar{\xi})}{\partial\xi}\right\|^2\quad(\text{动力学约束})$$

(**c**)轨迹损失函数：

$$L_{trajectory} = \left\| \xi_{t+\Delta t} - Integrate(\xi_t, \xi_{predict}, \Delta t) \right\|^2 \quad \text{(单步积分残差约束)}$$

其中

$$\begin{cases} \xi_{predict} = -i \dfrac{\partial \widetilde{H}_\lambda(\xi, \bar{\xi})}{\partial \bar{\xi}}|_{\xi=\xi_t}, \xi_t = \dfrac{1}{\sqrt{2}}(q_t + ip_t) \in \mathbb{C}^N \quad \text{(可直接观测的量)} \\ Integrate: RK4\text{(可能更优)或 } Euler \text{ 积分器等尝试, 步长(}\Delta t\text{)小，可调节} \end{cases}$$

也可增加其它损失函数进去，待定

物理机理与数据融合训练 Part 4

(ii) 复值激活函数

•幅角保持激活函数$modReLU(z)$

$$modReLU(z) = ReLU(|z| + b)e^{i\theta_z} = \begin{cases} (|z| + b)\frac{z}{|z|}, & |z| + b > 0 \\ 0, & \text{其它} \end{cases}$$ 不解析

•完全复数激活函数$zReLU(z)$

$$zReLU(z) = \begin{cases} z, & \theta_z \in [0, \frac{\pi}{2}] \\ 0, & \text{其它} \end{cases}$$ 在 x, y 正半轴不解析。

•Split 激活函数$CReLU(z)$

$CReLU(z) = ReLU(Re(z)) + i\,ReLU(Im(z))$ 在第一、三象限解析

小结：复激活函数分三类（以此为例）：
1. Split 激活函数（分离实部与虚部的激活函数）$CReLU(z)$
2. 完全复数激活函数$zReLU(z)$
3. 幅角保持激活函数$modReLU(z)$

刘维尔定理：在复平面上**处处解析**且**有界**的函数必是常数。

因此，在设计复值神经网络时，①“处处解析”与②“有界”不能同时成立（否则只能是常数函数，没有实用价值）。

注：解析函数类由定义域上每一点都可导的复变函数组成。

(iii) 解析条件与柯西-黎曼方程组推导

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$$

$$f(z) := u(z) + iv(z)$$

令 h 为实数，则 $f'(z) = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$ (式 a)

取 h 为纯虚数 ik，则$f'(z) = \lim_{k \to 0} \frac{f(z+ik) - f(z)}{ik} = \lim_{k \to 0} \frac{f(x+iy+ik) - f(x+iy)}{ik} = \lim_{k \to 0} \frac{f(x+i(y+k)) - f(x+iy)}{ik} =$

$-i \lim_{k \to 0} \frac{f(x+i(y+k)) - f(x+iy)}{k} = -i\frac{\partial f}{\partial y} = -i(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y})$(式 b)

联立式 a 与式 b：

要使 $f'(z)$存在且唯一，两式必须相等：

$$\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x}=\frac{\partial v}{\partial y}-i\frac{\partial u}{\partial y}$$

分别令实部与虚部相等，即得 柯西-黎曼方程组：

$$\begin{cases}\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y}\\ \frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}\end{cases}(27)$$

此即复变函数 $f(z)$ 解析的必要条件。

$\frac{\partial f}{\partial x}=-i\frac{\partial f}{\partial y}$(28)

有(a)知

$$f'(z)=\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x}$$

$$|f'(z)|^2=\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial v}{\partial x}\right)^2\overset{(27)}{=}\frac{\partial u}{\partial x}\left(\frac{\partial v}{\partial y}\right)+\frac{\partial v}{\partial x}\left(-\frac{\partial u}{\partial y}\right)=\frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial u}{\partial y}\frac{\partial v}{\partial x}=\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}$$

u,v是关于 x,y 的雅可比行列式

$$\Delta u=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\overset{(27)}{=}\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)=\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right)+\frac{\partial}{\partial y}\left(-\frac{\partial v}{\partial x}\right)=\frac{\partial^2 v}{\partial y\partial x}-\frac{\partial^2 v}{\partial x\partial y}=0$$

$(u,v$有各阶连续偏导数)

$\Delta u=0$:u 是调和函数,

$\Delta v=0$: v是调和函数,$-v(x,y+k)+v(x,y+k)$

u,v为共轭调和函数

若u,v为一对共轭调和函数，令

$$f(z)=u+iv,\ \begin{cases}u=u(x,y)\\ v=v(x,y)\end{cases}\text{,则}$$

$$\lim_{h+ik\to 0}\frac{f(z+h+ik)-f(z)}{h+ik}=\lim_{h+ik\to 0}\frac{[u(x+h,y+k)+iv(x+h,y+k)]-[u(x,y)+iv(x,y)]}{h+ik}$$

$$=\lim_{h+ik\to 0}\frac{[u(x+h,y+k)-u(x,y+k)+u(x,y+k)-u(x,y)]+i[v(x+h,y+k)-v(x,y+k)+v(x,y+k)-v}{h+ik}$$

$$=\lim_{h+ik\to 0}\frac{1}{h+ik}\{[u(x+h,y+k)-u(x,y+k)+u(x,y+k)-u(x,y)]+$$

$i[v(x+h,y+k)-v(x,y+k)+v(x,y+k)-v(x,y)]\}$

$$=\lim_{h+ik\to 0}\frac{(\frac{\partial u}{\partial x}h+\frac{\partial u}{\partial y}k)+\varepsilon_1+i(\frac{\partial v}{\partial x}h+\frac{\partial v}{\partial y}k+\varepsilon_2)}{h+ik}$$

$$=\lim_{h+ik\to 0}\frac{(\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x})h+(\frac{\partial u}{\partial y}+i\frac{\partial v}{\partial y})k+(\varepsilon_1+i\varepsilon_2)}{h+ik}$$

$$\overset{(27)}{=}\lim_{h+ik\to 0}\frac{(\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x})h+(-\frac{\partial v}{\partial x}+i\frac{\partial u}{\partial x})k+(\varepsilon_1+i\varepsilon_2)}{h+ik}$$

$$=\lim_{h+ik\to 0}\frac{(\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x})h+i(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial x})k+(\varepsilon_1+i\varepsilon_2)}{h+ik}$$

$$= \lim_{h+ik\to 0} \frac{(\frac{\partial u}{\partial x} + i\frac{\partial u}{\partial x})(h+ik) + (\varepsilon_1 + i\varepsilon_2)}{h+ik}, \begin{cases} \varepsilon_1 = \mathrm{o}(h+ik) \\ \varepsilon_2 = \mathrm{o}(h+ik) \end{cases}$$

$$= \frac{\partial u}{\partial x} + i\frac{\partial u}{\partial x}$$

$$\Rightarrow f'(z) \text{ 解析}$$

$Thm.\, u(x,y), v(x,y)$满足$(Cauchy - Riemann$方程$)$，有连续一阶偏导数

$\Leftrightarrow f(z) = u(x,y) + iv(x,y)$ 解析, $f'(z)$ 连续.

$$f(x,y),x,y\text{实变量}.f(x,y)\in\mathbb{C}$$

$$\begin{cases}z=\mathrm{x}+\mathrm{iy}\\ \overline{z}=x-iy\end{cases}\Rightarrow\begin{cases}x=\dfrac{1}{2}(z+\overline{z})\\ y=\dfrac{1}{2i}(z-\overline{z})=-\dfrac{i}{2}(z-\overline{z})\end{cases}$$

$$f(x,y)=f(\frac{1}{2}(z+\overline{z}),-\frac{i}{2}(z-\overline{z}))\text{是}z,\overline{z}\text{的函数}$$

$$\frac{\partial f}{\partial z}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial z}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial z}=\frac{1}{2}\frac{\partial f}{\partial x}-\frac{i}{2}\frac{\partial f}{\partial y}=\frac{1}{2}\left(\frac{\partial f}{\partial x}-i\frac{\partial f}{\partial y}\right)$$

$$\frac{\partial f}{\partial \overline{z}}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial \overline{z}}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial \overline{z}}=\frac{1}{2}\frac{\partial f}{\partial x}+\frac{i}{2}\frac{\partial f}{\partial y}=\frac{1}{2}\left(\frac{\partial f}{\partial x}+i\frac{\partial f}{\partial y}\right)$$

$$f\text{解析}\Rightarrow(28)\text{成立}\Rightarrow\frac{\partial f}{\partial \overline{z}}=0$$

哈密顿方程是 Binkhoff 方程的退化形式

Binkhoff 方程一般形式：

$$\left(\frac{\partial F_j}{\partial z_i}-\frac{\partial F_i}{\partial z_j}\right)\dot{z}_j-\left(\frac{\partial B}{\partial z_i}+\frac{\partial F_i}{\partial t}\right)=0,\quad i=1,2,\cdots,2n\quad(\bigstar)$$

$$B=B(z,t):Birkhoff\text{函数}\quad z=(z_1,\cdots,z_n,p_1,\cdots,p_n)$$

当取

$$\begin{cases}z_i=\begin{cases}q_i, & i=1,2,\cdots,n\\ p_{i-n}, & i=n+1,n+2,\cdots,2n\end{cases}\\ F_i=\begin{cases}p_i, & i=1,2,\cdots,n\\ 0, & i=n+1,n+2,\cdots,2n\end{cases}\\ B=H\end{cases}$$

时，Birkhoff 方程退化为哈密顿方程

$$F_i=F_i(z,t)$$

$$\mathrm{n}=1: F_i(q,p,t)\text{三元函数}$$

$$e.g.n=1$$

$$z_i=\begin{cases}q_1\\p_1\end{cases},F_i=\begin{cases}p_1,i=1\\0,i=2\end{cases}\Rightarrow z_1=q_1\triangleq q,z_2=p_1\triangleq p,F_1=p_1\triangleq p,F_2=0$$

$$\begin{cases}\left(\dfrac{\partial F_2}{\partial z_1}-\dfrac{\partial F_1}{\partial z_2}\right)\dot{z}_2-\dfrac{\partial B}{\partial z_1}-\dfrac{\partial F_1}{\partial t}=0\\ \left(\dfrac{\partial F_1}{\partial z_2}-\dfrac{\partial F_2}{\partial z_1}\right)\dot{z}_1-\dfrac{\partial B}{\partial z_2}-\dfrac{\partial F_2}{\partial t}=0\end{cases}$$

代入设定值:

$$\begin{cases}\left(0-\dfrac{\partial p}{\partial p}\right)\dot{p}-\dfrac{\partial H}{\partial q}-\dfrac{\partial p}{\partial t}=0\\ \left(\dfrac{\partial p}{\partial p}-0\right)\dot{q}-\dfrac{\partial H}{\partial p}-0=0\end{cases}$$

最终得到经典的哈密顿正则方程：

$$\begin{cases}\dot{q}=\dfrac{\partial H}{\partial p}\\ \dot{p}=-\dfrac{\partial H}{\partial q}\end{cases}$$

回顾方程(12)$\begin{cases}\dot{q}=\frac{\partial H}{\partial q}=M^{-1}(q)p\text{ , (a)}\\ \dot{p}=-\frac{\partial H}{\partial q}+F_{ext}-C\frac{\partial H}{\partial p}=-\frac{\partial H}{\partial q}+F_{ext}-CM^{-1}p\text{, (b)}\end{cases}$

$$\mathrm{z}=(q,p)^T,\text{则}$$

$$(\bigstar)\Rightarrow\begin{cases}\left(\dfrac{\partial F_2}{\partial \mathrm{q}}-\dfrac{\partial F_1}{\partial \mathrm{p}}\right)\dot{p}-\left(\dfrac{\partial B}{\partial \mathrm{q}}+\dfrac{\partial F_1}{\partial t}\right)\overset{F_2=0}{\Longrightarrow}-\dfrac{\partial F_1}{\partial \mathrm{p}}\dot{p}=\dfrac{\partial B}{\partial \mathrm{q}}+\dfrac{\partial F_1}{\partial t},(\mathrm{c})\\ \left(\dfrac{\partial F_1}{\partial \mathrm{p}}-\dfrac{\partial F_2}{\partial \mathrm{q}}\right)\dot{q}-\left(\dfrac{\partial B}{\partial \mathrm{p}}+\dfrac{\partial F_2}{\partial t}\right)\overset{F_2=0}{\Longrightarrow}\dfrac{\partial F_1}{\partial \mathrm{p}}\dot{q}=\dfrac{\partial B}{\partial \mathrm{p}},(\mathrm{d})\end{cases}$$

$$F_1(\mathrm{q},\mathrm{p}),\text{假设}\begin{cases}F_1\text{里全含 t}\\ \dfrac{\partial F_1}{\partial \mathrm{p}}=1\end{cases},\text{则(c),(d)}\Rightarrow\begin{cases}\dot{p}=-\dfrac{\partial B}{\partial \mathrm{q}}-\dfrac{\partial F_1}{\partial t},(\mathrm{e})\\ q=\dfrac{\partial B}{\partial \mathrm{p}},(\mathrm{f})\end{cases}$$

$$\text{(e),(f)与(a),(b)对照，可取}\begin{cases}\dfrac{\partial B}{\partial p}=\dfrac{\partial H}{\partial p}=M^{-1}(q)p\\ \dfrac{\partial B}{\partial \mathrm{q}}+\dfrac{\partial F_1}{\partial t}=\dfrac{\partial H}{\partial q}+(CM^{-1}p-F_{ext})\end{cases}$$

$$\begin{cases}B=H=\dfrac{1}{2}\mathrm{p}^TM\mathrm{q}+\int \mathrm{K}(q)q\,\mathrm{d}q\\ F_1=\int[\mathrm{C}M^{-1}\mathrm{q}-F_{ext}(q)]\,\mathrm{d}t+p\end{cases}$$

即有(12)可以转化为

$$\begin{cases}(\dfrac{\partial F_2}{\partial q}-\dfrac{\partial F_1}{\partial p})\dot{p}=\dfrac{\partial B}{\partial q}+\dfrac{\partial F_1}{\partial t}\\ (\dfrac{\partial F_1}{\partial p}-\dfrac{\partial F_2}{\partial q})\dot{q}=\dfrac{\partial B}{\partial q}\end{cases}$$

其中$\begin{cases}B=\frac{1}{2}\mathrm{p}^TM\mathrm{q}+\int \mathrm{K}(q)q\,\mathrm{d}q\\ F_1=\int[\mathrm{C}M^{-1}\mathrm{q}-F_{ext}(q)]\,\mathrm{d}t+p\\ F_2=0\end{cases}$

构造生成函数ω，可得 Birkhoff 保辛格式

需要用到两个定理

$$\text{记}p=\begin{pmatrix}\hat{z}\\ z\end{pmatrix}\in\mathbb{R}^{4n},\ z=z(t_0),\ \hat{z}=\hat{z}(t)$$

$$\alpha(t_1t_0):\begin{pmatrix}\hat{z}\\ z\end{pmatrix}\mapsto\begin{pmatrix}\hat{w}\\ w\end{pmatrix}=\begin{pmatrix}\alpha_1(\hat{z},z,\mathrm{t},t_0)\\ \alpha_2(\hat{z},z,\mathrm{t},t_0)\end{pmatrix}$$

$$\alpha^{-1}(t_2t_0):\begin{pmatrix}\hat{w}\\ w\end{pmatrix}\mapsto\begin{pmatrix}\hat{z}\\ z\end{pmatrix}=\begin{pmatrix}\alpha^1(\hat{w},w,\mathrm{t},t_0)\\ \alpha^2(\hat{w},w,\mathrm{t},t_0)\end{pmatrix}$$

$$\alpha，\ \alpha^{-1}\text{的}Jacobi\text{矩阵为}$$

$\alpha_?=\begin{pmatrix}A_\alpha & B_\alpha\\ C_\alpha & D_\alpha\end{pmatrix}$，？

$\alpha_?^{-1}=\begin{pmatrix}A^\alpha & B^\alpha\\ C^\alpha & D^\alpha\end{pmatrix}$。？

则有

Theorem (A)

对上述映射α，映射 $\mathrm{z}\mapsto\hat{z}=g(t;\alpha,t_0)$是 Birkhoff 系统的相流，

$\mathrm{M}(t;z,t_0)=g_z(t;z,t_0)$是其 Jacobi 矩阵。
若其行列式满足

$$|C_\alpha(z,z,t_0,t_0)+D_\alpha(z,z,t_0,t_0)|\neq 0$$

则当$|t-t_0|$足够小时，在z的某邻域内存在梯度映射：

$$w\to\hat{w}=f(w,t,t_0)$$

此映射的雅可比矩阵$f_w(w,t,t_0)=N(w,t,t_0)$对称，且存在生成函数$\phi(w,t,t_0)$，使得：

$$f(w,t,t_0)=\phi_w(w,t,t_0)$$

$$\frac{\partial}{\partial t}\phi_w(w,t,t_0)=A[\phi_w(w,t,t_0);\ w,\ \phi_{ww}(w,t,t_0),\ t,\ t_0]$$

$$A(\hat{w},w,\frac{\partial\hat{w}}{\partial w},t,t_0)=\overline{A}\left[\hat{z}(\hat{w},w,t,t_0);z(\hat{w},w,t,t_0),\ \frac{\partial\hat{w}}{\partial w},\ t,\ t_0\right]$$

$$\overline{A}(\hat{z},z,\frac{\partial\hat{w}}{\partial w},t,t_0)=\frac{d}{dt}\hat{w}(\hat{z},z,t,t_0)-\frac{\partial\hat{w}}{\partial w}\frac{d}{dt}w(\hat{z},z,t,t_0)$$

$$=\left(A_\alpha-\frac{\partial\hat{w}}{\partial w}C_\alpha\right)K^{-1}D(\hat{z},t)+\frac{\partial\alpha_1}{\partial t}-\frac{\partial\hat{w}}{\partial w}\frac{\partial\alpha_2}{\partial t}$$

$$\alpha_1[\mathrm{g}(t,z,t_0),z,t,t_0]=f\{\alpha_2[g(t;\alpha,t_0),z,t,t_0],\ t,\ t_0\}=\phi_w[\alpha_2(\mathrm{g}(t;\alpha,t_0),z,t,t_0),\ t,\ t_0]$$

α_1关于(z,t_0)唯一且

$$N=\sigma_\alpha(M)=(A_\alpha M+B_\alpha)(C_\alpha M+D_\alpha)^{-1}$$
$$M=\sigma_{\alpha^{-1}}(N)=(A^\alpha N+B^\alpha)(C^\alpha N+D^\alpha)^{-1}$$

Theorem (B)

1. 解析性与级数展开：

假设A,α解析，则当$|t-t_0|$足够小时，生成函数$\phi_w(w,t,t_0)$可对t展开成收敛级数

$$\phi_w(w,t,t_0)=\sum_{k=0}^{+\infty}(t-t_0)^k\phi_w^{(k)}(w,t_0)$$

$$\phi_w^{(0)}(w,t_0)=f(w,t_0,t_0)$$

$$\phi_w^{(1)}(w,t_0)=A\left(\phi_w^{(0)},\ w,\ \phi_{ww}^{(0)},\ t_0,\ t_0\right)$$

$$\phi_w^{(k)}(w,t_0)=\frac{1}{(k+1)!}D_t^kA\left(\phi_w^{(0)},\ w,\ \phi_{ww}^{(0)},\ t_0,\ t_0\right)$$

构造

$$\begin{cases}\hat{q}(t)=q(t+\delta)\\ \hat{p}(t)=p(t+\delta)\\ \widehat{Q}=\hat{p}-p\\ \widehat{P}=\hat{q}-q\\ Q=\frac{1}{2}(\hat{q}+q)\\ P=-\frac{1}{2}(\hat{p}+p)\end{cases}$$ (30)“左大右小”

变量定义：

$$\hat{z}=\begin{pmatrix}\hat{q}\\ \hat{p}\end{pmatrix},\quad z=\begin{pmatrix}q\\ p\end{pmatrix},\quad \widehat{w}=\begin{pmatrix}\widehat{Q}\\ \widehat{P}\end{pmatrix},\quad w=\begin{pmatrix}Q\\ P\end{pmatrix}$$

变换α：

$$\alpha:\begin{pmatrix}\hat{q}\\ \hat{p}\\ q\\ p\end{pmatrix}=\begin{pmatrix}\hat{z}\\ z\end{pmatrix}\mapsto\begin{pmatrix}\widehat{w}\\ w\end{pmatrix}=\begin{pmatrix}\widehat{Q}\\ \widehat{P}\\ Q\\ P\end{pmatrix}$$

α的 *Jacobi* 矩阵α_*（分块矩阵）：

$$\alpha_*=\begin{pmatrix}\frac{\partial\widehat{Q}}{\partial\hat{q}} & \frac{\partial\widehat{Q}}{\partial\hat{p}} & \frac{\partial\widehat{Q}}{\partial q} & \frac{\partial\widehat{Q}}{\partial p}\\ \frac{\partial\widehat{P}}{\partial\hat{q}} & \frac{\partial\widehat{P}}{\partial\hat{p}} & \frac{\partial\widehat{P}}{\partial q} & \frac{\partial\widehat{P}}{\partial p}\\ \frac{\partial Q}{\partial\hat{q}} & \frac{\partial Q}{\partial\hat{p}} & \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p}\\ \frac{\partial P}{\partial\hat{q}} & \frac{\partial P}{\partial\hat{p}} & \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p}\end{pmatrix}=\begin{pmatrix}0 & I & 0 & -I\\ I & 0 & -I & 0\\ \frac{1}{2}I & 0 & \frac{1}{2}I & 0\\ 0 & -\frac{1}{2}I & 0 & -\frac{1}{2}I\end{pmatrix}$$

由(30)知(31)“左小右大”

$$\begin{cases}\hat{q}=\dfrac{\widehat{P}+2Q}{2}=Q+\dfrac{1}{2}\widehat{P}\\ \hat{p}=\dfrac{\widehat{Q}-2P}{2}=\dfrac{1}{2}\widehat{Q}-P\\ q=\dfrac{2Q-\widehat{P}}{2}=Q-\dfrac{1}{2}\widehat{P}\\ p=\dfrac{-2P-\widehat{Q}}{2}=-\dfrac{1}{2}\widehat{Q}-P\end{cases}$$

变换α^{-1}：

$$\alpha^{-1}:\begin{pmatrix}\widehat{Q}\\ \widehat{P}\\ Q\\ P\end{pmatrix}=\begin{pmatrix}\widehat{w}\\ w\end{pmatrix}\mapsto\begin{pmatrix}\hat{z}\\ z\end{pmatrix}=\begin{pmatrix}\hat{q}\\ \hat{p}\\ q\\ p\end{pmatrix}$$

α^{-1}的$Jacobi$矩阵α_*^{-1}（分块矩阵）：

$$\alpha_*^{-1}=\begin{pmatrix}0 & \frac{1}{2}I & I & 0\\ \frac{1}{2}I & 0 & 0 & -I\\ 0 & -\frac{1}{2}I & I & 0\\ -\frac{1}{2}I & 0 & 0 & -I\end{pmatrix}$$

$$\frac{d\widehat{w}}{dt}=\begin{pmatrix}0 & I\\ I & 0\end{pmatrix}\begin{pmatrix}\dot{\hat{q}}\\ \dot{\hat{p}}\end{pmatrix}=\begin{pmatrix}\dot{\hat{p}}\\ \dot{\hat{q}}\end{pmatrix}\overset{\text{Page6}}{=}\begin{pmatrix}-\frac{1}{2}\hat{p}^T\frac{\partial M^{-1}(\hat{q})}{\partial\hat{q}}\hat{p}-K(\hat{q})\hat{q}+F_{ext}-CM^{-1}\hat{p}\\ M^{-1}(\hat{q})\hat{p}\end{pmatrix}$$

$$\overset{(31)}{=}\begin{pmatrix}-\frac{1}{2}\left(\frac{1}{2}\widehat{Q}-P\right)^T\frac{\partial M^{-1}(\mathrm{Q}+\frac{1}{2}\widehat{P})}{\partial(\mathrm{Q}+\frac{1}{2}\widehat{P})}\left(\frac{1}{2}\widehat{Q}-p\right)-K\left(\widehat{Q}+\frac{1}{2}\widehat{P}\right)\left(\widehat{Q}+\frac{1}{2}\widehat{P}\right)+F_{ext}-CM^{-1}\left(\frac{1}{2}\widehat{Q}-P\right)\\ M^{-1}\left(Q+\frac{1}{2}\widehat{P}\right)\left(\frac{1}{2}\widehat{Q}-P\right)\end{pmatrix}$$

$$\frac{dw}{dt}=\begin{pmatrix}\frac{1}{2}I & 0\\ 0 & -\frac{1}{2}I\end{pmatrix}\begin{pmatrix}\dot{q}\\ \dot{p}\end{pmatrix}=\begin{pmatrix}\frac{1}{2}\dot{q}\\ \frac{1}{2}\dot{p}\end{pmatrix}=\begin{pmatrix}\frac{1}{2}[-\frac{1}{2}\hat{p}^T\frac{\partial M^{-1}(\hat{q})}{\partial\hat{q}}\hat{p}-K(\hat{q})\hat{q}+F_{ext}-CM^{-1}\hat{p}]\\ \frac{1}{2}M^{-1}(\hat{q})\hat{p}\end{pmatrix}\Big|_{\begin{cases}p=-\frac{1}{2}\widehat{Q}-P\\ q=Q-\frac{1}{2}\widehat{P}\end{cases}}(32$$

)

$$\frac{d\widehat{w}}{dt}=\begin{pmatrix}0 & I\\ I & 0\end{pmatrix}\begin{pmatrix}\dot{\hat{q}}\\ \dot{\hat{p}}\end{pmatrix}=\begin{pmatrix}\dot{\hat{p}}\\ \dot{\hat{q}}\end{pmatrix}=\begin{pmatrix}-\frac{1}{2}\hat{p}^T\frac{\partial M^{-1}(\hat{q})}{\partial\hat{q}}\hat{p}-K(\hat{q})\hat{q}+F_{ext}-CM^{-1}\hat{p}\\ M^{-1}(\hat{q})\hat{p}\end{pmatrix}\Big|_{\begin{cases}\hat{p}=\frac{1}{2}\widehat{Q}-P\\ \hat{q}=Q+\frac{1}{2}\widehat{P}\end{cases}}$$

由 Theorem(B)知

$$\phi_w(w,t,t_0)=\phi_w^{(0)}(w,t_0)+\phi_w^{(1)}(w,t_0)(t-t_0)+\phi_w^{(2)}(w,t_0)(t-t_0)^2+\cdots$$

$$\phi_w^{(0)}(w,t_0)=\hat{w}|_{t=t_0}=\left.\begin{pmatrix}\hat{Q}\\ \hat{P}\end{pmatrix}\right|_{t=t_0}=\left.\begin{pmatrix}\hat{Q}\\ \hat{P}\end{pmatrix}\right|_{\delta=0}=\begin{pmatrix}0\\0\end{pmatrix}$$

$$\phi_w^{(1)}(w,t_0)=A(\phi_w^{(0)},w,\phi_{ww}^{(0)},t_0,t_0)=A(\hat{w}|_{t=t_0},w,\left.\frac{\partial\hat{w}}{\partial \mathrm{w}}\right|_{t=t_0},t_0,t_0)=\left.\frac{d\hat{w}}{d\mathrm{t}}\right|_{t=t_0}-\left.\frac{\partial\hat{w}}{\partial \mathrm{w}}\right|_{t=t_0}\left.\frac{dw}{dt}\right|_{t=t_0}$$

$$\left.\frac{d\hat{w}}{d\mathrm{t}}\right|_{t=t_0}\overset{(32)}{=}\left.\frac{d\hat{w}}{d\mathrm{t}}\right|_{\delta=0}=\begin{pmatrix}-\frac{1}{2}\mathrm{P}^T\frac{\partial M^{-1}}{\partial \mathrm{Q}}\mathrm{P}-K(\mathrm{Q})\mathrm{Q}+F_{ext}-CM^{-1}\mathrm{P}\\ -M^{-1}(\mathrm{Q})\mathrm{P}\end{pmatrix}$$

当$\delta=0$ 时，由(30),(31)知$\begin{cases}\hat{q}=q,\hat{p}=p,\hat{Q}=0,\hat{P}=0,Q=q,P=-p\\ \Rightarrow\hat{p}=-\mathrm{P},\hat{q}=\mathrm{Q}\end{cases}$

$$\left.\frac{d\hat{w}}{d\mathrm{w}}\right|_{t=t_0}=\frac{\partial(\hat{w}|_{t=t_0})}{\partial \mathrm{w}}=\frac{\partial(0)}{\partial \mathrm{w}}=0$$

$$\Rightarrow\phi_w^{(1)}(w,t_0)=\begin{pmatrix}-\frac{1}{2}\mathrm{P}^T\frac{\partial M^{-1}}{\partial \mathrm{Q}}\mathrm{P}-K(\mathrm{Q})\mathrm{Q}+F_{ext}-CM^{-1}\mathrm{P}\\ -M^{-1}(\mathrm{Q})\mathrm{P}\end{pmatrix},\begin{cases}C=C(Q,\dot{Q})\\ M=M(Q)\end{cases}$$

$$\overset{Thm(A),Thm(B)}{\Longrightarrow}\hat{w}=\phi_w^{(0)}(w,t_0)+\phi_w^{(1)}(w,t_0)(t-t_0)+\mathrm{o}(2)=\phi_w^{(1)}(w,t_0)(t-t_0),t-t_0=\delta(33)$$

记$\begin{cases}\hat{p}=p_{\mathrm{k+1}},p=p_{\mathrm{k}}\\ \hat{q}=q_{\mathrm{k+1}},q=q_{\mathrm{k}}\end{cases}$

$$\begin{cases}P=-\frac{1}{2}(\hat{p}+p)\\ Q=\frac{1}{2}(\hat{q}+q)\end{cases},\hat{w}=\phi_w^{(1)}(w,t_0)\delta=\begin{pmatrix}\hat{Q}\\ \hat{P}\end{pmatrix}=\begin{pmatrix}\hat{p}-p\\ \hat{q}-q\end{pmatrix}(34)$$

由(33)(34)有

$$\begin{pmatrix}\hat{p}-p\\ \hat{q}-q\end{pmatrix}=\left(\begin{matrix}-\frac{1}{2}\mathrm{P}^{T}\frac{\partial M^{-1}}{\partial \mathrm{Q}}\mathrm{P}-K(\mathrm{Q})\mathrm{Q}+F_{ext}-CM^{-1}\mathrm{P}\\ -M^{-1}(\mathrm{Q})\mathrm{P}\end{matrix}\right)\Bigg|_{\substack{P=-\frac{1}{2}(\hat{p}+p)\\ Q=\frac{1}{2}(\hat{q}+q)}}^{\delta}$$

$$=\begin{pmatrix}-\frac{1}{2}[-\frac{1}{2}(\hat{p}+p)]^{T}\frac{\partial M^{-1}}{\partial\left\{\frac{1}{2}(\hat{q}+q)\right\}}[-\frac{1}{2}(\hat{p}+p)]-K\left\{\frac{1}{2}(\hat{q}+q)\right\}\frac{1}{2}(\hat{q}+q)+F_{ext}-CM^{-1}[-\frac{1}{2}(\hat{p}+p)]\\ -M^{-1}\left\{\frac{1}{2}(\hat{q}+q)\right\}[-\frac{1}{2}(\hat{p}+p)]\end{pmatrix}$$

$$\Rightarrow\begin{cases}\hat{q}=q+\frac{1}{2}M^{-1}\{\frac{1}{2}(\hat{q}+q)\}(\hat{p}+p)\cdot\delta\\ \hat{p}=p+\left\{-\frac{1}{8}(\hat{p}+p)\frac{\partial M^{-1}}{\partial\left\{\frac{1}{2}(\hat{q}+q)\right\}}(\hat{p}+p)-\frac{1}{2}K\{(\hat{q}+q)\}(\hat{q}+q)+F_{ext}-\frac{1}{2}CM^{-1}(\hat{p}+p)\right\}\delta\end{cases}$$

简记为

$$\Rightarrow\begin{cases}\hat{q}=q+\frac{1}{2}M^{-1}(\hat{p}+p)\cdot\delta\\ \hat{p}=p+\left\{-\frac{1}{8}(\hat{p}+p)\frac{\partial M^{-1}}{\partial\left\{\frac{1}{2}(\hat{q}+q)\right\}}(\hat{p}+p)-\frac{1}{2}K(\hat{q}+q)+F_{ext}-\frac{1}{2}CM^{-1}(\hat{p}+p)\right\}\delta\end{cases}\quad(35)$$

将(35)第一个方程代入第二个方程，有

$$\hat{p}=p+\left\{-\frac{1}{8}(\hat{p}+p)\frac{\partial M^{-1}}{\partial\left\{\frac{1}{2}(\hat{q}+q)\right\}}(\hat{p}+p)-\frac{1}{2}K(2q+\frac{1}{2}M^{-1}(\hat{p}+p)\cdot\delta)+F_{ext}-\frac{1}{2}CM^{-1}(\hat{p}+p)\right\}\delta$$

移项得

$$\hat{p}(1+\frac{1}{4}KM^{-1}\delta^{2}+\frac{1}{2}CM^{-1}\delta)=p(1-\frac{1}{4}KM^{-1}\delta^{2}-\frac{1}{2}CM^{-1}\delta)-Kq\cdot\delta+(\Delta+F_{ext})\delta$$

若 M，K，F_{ext} 为常矩阵，则$\Delta=0$, 记$\varepsilon=\frac{1}{4}KM^{-1}\delta^{2}+\frac{1}{2}CM^{-1}\delta$, 则

$$\hat{p} = \frac{1}{1+\varepsilon}[p_k(1-\varepsilon) - Kq_k\delta + F_{ext}\delta]$$

从而

$$\begin{cases} q_{k+1} = q_k + \dfrac{1}{2}M^{-1}(p_{k+1} + p_k)\cdot\delta \\ p_{k+1} = p_k \cdot \dfrac{1-\varepsilon}{1+\varepsilon} - Kq_k \cdot \dfrac{\delta}{1+\varepsilon} + F_{ext}\dfrac{\delta}{1+\varepsilon} \end{cases}$$

若M，K，$\mathrm{F_{ext}}$不是常矩阵时,即$M = M(Q)$，$K = K(Q,\dot{Q})$,则$\Delta \neq 0$，　Δ为二次型

$$\begin{cases} q_{k+1} = q_k + \frac{1}{2}M^{-1}(p_{k+1} + p_k)\cdot\delta, \text{其中 } M^{-1} \text{ 依赖于 } q_{k+1} + q_k \\ p_{k+1} = p_k \cdot \frac{1-\varepsilon}{1+\varepsilon} - Kq_k \cdot \frac{\delta}{1+\varepsilon} + (\Delta + F_{ext})\frac{\delta}{1+\varepsilon} \end{cases} \begin{cases} K, \mathrm{F_{ext}}\varepsilon, C\text{依赖于} q_{k+1} + q_k \\ \Delta \text{ 依赖于 } p_{k+1} = p_k, q_{k+1} + q_k \end{cases} \quad (36)$$

Birkhoff 保辛格式，一阶精度

这是一个隐式方法

$$\phi_w^{(2)}(w,t_0)\overset{Thm(B)}{=}\frac{1}{2}\mathrm{D_t}\phi_w^{(1)}(w,t_0)\overset{(33)}{=}\frac{1}{2}\mathrm{D_t}\begin{pmatrix}-\frac{1}{2}\mathrm{P}^T\frac{\partial M^{-1}}{\partial \mathrm{Q}}\mathrm{P}-K(\mathrm{Q})\mathrm{Q}+F_{ext}+CM^{-1}\mathrm{P}\\-M^{-1}(\mathrm{Q})\mathrm{P}\end{pmatrix}$$

已知

$$\begin{cases}\dot{Q}=-M^{-1}\mathrm{P}\\ \dot{\mathrm{P}}=\mathrm{KQ}-F_{ext}-CM^{-1}\mathrm{P}\end{cases}$$

在(30)中

$$\begin{cases}Q=\frac{1}{2}(\hat{q}+q)\\ P=-\frac{1}{2}(\hat{p}+p)\end{cases}\begin{cases}\hat{q}(t)=q(t+\delta)\\ \hat{p}(t)=p(t+\delta)\end{cases}$$

，显然可以看出来

事实上$\begin{cases}\dot{q}=M^{-1}(\mathrm{q})\mathrm{p}\\ \dot{p}=-K(\mathrm{q})\mathrm{q}+F_{ext}(q)-C(q,\dot{q})M^{-1}(\mathrm{q})\mathrm{p}\end{cases}$

$\begin{cases}\dot{\hat{q}}=M^{-1}(\hat{q})\hat{p}\\ \dot{\hat{p}}=-K(\hat{q})\hat{q}+F_{ext}(\hat{q})-C(\hat{q},\dot{\hat{q}})M^{-1}(\mathrm{q})\hat{p}\end{cases}$

$$\Rightarrow\dot{Q}=\frac{1}{2}(\dot{\hat{q}}+\dot{q})=\frac{1}{2}(M^{-1}(\mathrm{q})\mathrm{p}+M^{-1}(\hat{q})\hat{p})$$

若 M 为常矩阵，则$\dot{Q}=M^{-1}\frac{1}{2}(\mathrm{p}+\hat{p})=-M^{-1}\mathrm{p}$

$$\dot{P}=-\frac{1}{2}(\dot{\hat{p}}+\dot{p})=$$

$$-\frac{1}{2}\{[-K(\hat{q})\hat{q}+F_{ext}(\hat{q})-C(\hat{q},\dot{\hat{q}})M^{-1}(\mathrm{q})\hat{p}]$$

$$+[-K(\mathrm{q})\mathrm{q}+F_{ext}(q)-C(q,\dot{q})M^{-1}(\mathrm{q})\mathrm{p}]\}$$

若F_{ext}， M，C 为常矩阵，则

$$\dot{P}=-\frac{1}{2}K(q+\hat{q})-F_{ext}+\frac{1}{2}CM^{-1}(\mathrm{p}+\hat{p})=KQ-F_{ext}-CM^{-1}\mathrm{P}$$

从而

$$\phi_w^{(2)}(w,t_0)=\frac{1}{2}\mathrm{D_t}\begin{pmatrix}0-KQ+F_{ext}+CM^{-1}P\\-M^{-1}P\end{pmatrix}=\frac{1}{2}\mathrm{D_t}\begin{pmatrix}0-K\dot{Q}+\dot{F_{ext}}+CM^{-1}\dot{P}\\-M^{-1}\dot{P}\end{pmatrix}$$
$$=\frac{1}{2}\mathrm{D_t}\begin{pmatrix}0-K(-M^{-1}p)+\dot{F_{ext}}+CM^{-1}(KQ-F_{ext}-CM^{-1}P)\\-M^{-1}(KQ-F_{ext}-CM^{-1}P)\end{pmatrix}$$
$$=\frac{1}{2}\mathrm{D_t}\begin{pmatrix}KM^{-1}p+\dot{F_{ext}}+CM^{-1}KQ-CM^{-1}F_{ext}-(CM^{-1})^2P\\-M^{-1}KQ+M^{-1}F_{ext}+M^{-1}CM^{-1}P\end{pmatrix}$$

用$Theorem(B)$的迭代公式，可以推出$\phi_w(w,t_0)$的形式$(k\geq 2)$. 从而分析生成函数$\hat{\omega}$

$=\phi_w(w,t,t_0)=\sum_{k=0}^{\infty}(t-t_0)^k\phi_w^{(k)}(w,t_0)$的截取形式. $\Gamma_K\phi_w(w,t;t_0):=\sum_{k=0}^{K}(t-t_0)^k\phi_w^{(k)}(w,t_0)$，即可得到$K$阶精度的$Birkhoff$保辛格式(计算机操作).

对方程 (36) 的讨论

对方程(36)：令$N=1$，即q,p都是一维变量，此时(36)为平面上的映射。
理论上，由生成函数法将$Birkhoff$系统转化为离散系统是保辛的。
原始系统（11）中的耗散项$K(q,\dot{q})\dot{q}$等，驱动项F_{ext}在$Birkhoff$方程中已通过"伪保辛化"处理（耗散项不强）。优势：几何结构保持

两种数值检验方式

- 对方程 (36)取初始值 $(q_0,p_0)\in\mathbb{R}^2$，生成 $\{(q_k,p_k),k\in\mathbb{Z}^+\}$ 轨道数据(36)中$N=1$ 时：
- ①看是否有封闭的、类似环状的分布，验证环面结构；
- ②*看功率谱分析，对轨道做*
  FFT，看是否有离散的不可公度的频率峰，验证环面上运动的拟周期性。

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(36) 的本质

（36）本质上是：$z_{k+1}=\Phi_h(z_k)$ （保辛离散迭代，离散动力学统）生成函数还保证了其是辛同胚的（$Symplectomorphism$）。

KAM 定理的离散版本（从两篇中选定理引用）

引用文献：

"$A\ note\ on\ KAM\ theorem\ for\ symplectic\ diffeomorphisms$", $J.D.E.5$, 2000, $Shang, Z.J.$
"$KAM\ theorem\ for\ symplectic\ algorithms\ for\ Hamiltonian\ systems$", $Numersche\ Mathematik$, 1999, $Shang$,
*

"要求步长δ足够小"

辨识未建模动力学效应 — Part 4

在（36）方程右侧加入控制项$(u_k,U_k):=U_k$得到
$z_{k+1}=\Phi_h(z_k,u_k)$， 用 U_k 调节模型的误差。 U_k 是动态的
或
$z_{k+1}=\Phi_h(z_k)+U_k$， 看哪个更好？
（控制项U_k用于抵消残差）
或在生成函数$\phi_w(w,t,t_0)$（$Theorem\ (B)$）中加入一个控制项U_k。
U_k的形式可尝试（不要太大、有界、非常值、简单有代表性）
……（其它与机器人、太空跳跃、微扰动……相关的陈述、自由发挥。）

评估误差与 KAM 阈值关联 Part 5

回顾 Theorem(A)，Theorem(B) 内容，看以下 pdf：

Symplectic Geometric Algorithms for Hamiltonian Systems, Kang Feng, Mengzhao Qin, Springer.

（看 Chapter 15: Structure Preserving Schemes for Birkhoff Systems）

Theorem (A) 即为 Theorem 4.1

Theorem (B) 即为 Theorem 4.4

一些记号：P18 (⋆) 这个方程即为 P618. (2.1)

$$K_{ij}=\frac{\partial F_j}{\partial z_i}-\frac{\partial F_i}{\partial z_j},\quad K=(K_{ij})_{i,j=1,2,\dots,2n}\quad (*_1)$$

$$由\ (\star)\Rightarrow K_{ij}(z,t)\frac{dz_j}{dt}-\frac{\partial B}{\partial z_i}-\frac{\partial F_i}{\partial t}=0,$$

记$D_i(z,t)=-\frac{\partial B}{\partial z_i}-\frac{\partial F_i}{\partial t}$ $(*_2)$

则 $B=B(z,t),;F_i=F_i(z,t)$ $\begin{cases}(2.3)\\(2.4)\end{cases}$

若 $\det(K_{ij})(\widetilde{Re})\neq 0$，则(2.3)是正则的。

(2.4) 是自伴随的

$$\Leftrightarrow\begin{cases}K_{ij}+K_{ji}=0 & （反对称性）\\ \frac{\partial K_{ij}}{\partial z_k}+\frac{\partial K_{jk}}{\partial z_i}+\frac{\partial K_{ki}}{\partial z_j}=0 & （雅可比恒等式 \Leftrightarrow 即\ d\Omega=0）\\ \frac{\partial K_{ij}}{\partial t}=\frac{\partial D_i}{\partial z_j}-\frac{\partial D_j}{\partial z_i} & （时间相容性条件）\end{cases}\quad (2.5)$$

$\widetilde{Re}^*$：$\mathbb{R}\times T^*M$ 上的星形区域，T^*M：M 的余切空间，M 是 $2n$ 维流形

（Ω是闭形式）：

$$\begin{cases} \Omega = \sum_{i,j=1}^{2n} K_{ij}(z,t), dz_i \wedge dz_j \end{cases} \Rightarrow d\Omega = 0 \Leftrightarrow \Omega \text{ 是恰当 2 形式} \Rightarrow \Omega = d(F_i dz_i) \quad (2.5)$$

$$D_i \overset{(*_2)}{=} -\frac{\partial B_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j} \Rightarrow -\frac{\partial B}{\partial z_i} = D_i + \frac{\partial F_i}{\partial t} \Rightarrow \frac{\partial}{\partial z_j}\left(-\frac{\partial B}{\partial z_i}\right) = \frac{\partial}{\partial z_i}\left(\frac{\partial B}{-\partial z_j}\right) = \frac{\partial}{\partial z_i}\left(D_j + \frac{\partial F_j}{\partial t}\right)$$

$$\Rightarrow \frac{\partial}{\partial z_j}\left(D_i + \frac{\partial F_i}{\partial t}\right) = \frac{\partial}{\partial z_i}\left(D_j + \frac{\partial F_j}{\partial t}\right)$$

$$\Rightarrow \frac{\partial D_j}{\partial z_j} - \frac{\partial D_i}{\partial z_i} = \frac{\partial}{\partial z_i}\left(\frac{\partial F_j}{\partial t}\right) - \frac{\partial}{\partial z_j}\left(\frac{\partial F_i}{\partial t}\right) \overset{*_1}{=} \frac{\partial}{\partial t}\left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right) = \frac{\partial}{\partial t}(K_{ij}) (*)$$

$\Rightarrow$ (2.5) 第三个方程成立

(2.4) 自伴随 $\Leftrightarrow$ (2.5)成立

$*_1, *_2 \Rightarrow$ (2.5) 第三个方程成立

$*_1 \Rightarrow K_{ij} + K_{ji} = \left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right) + \left(\frac{\partial F_i}{\partial z_j} - \frac{\partial F_j}{\partial z_i}\right) = 0 \Rightarrow$ (2.5) 第一个方程成立

$$\frac{\partial K_{ij}}{\partial z_k} + \frac{\partial K_{jk}}{\partial z_i} + \frac{\partial K_{ki}}{\partial z_j} = \frac{\partial}{\partial z_k}\left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right) + \frac{\partial}{\partial z_i}\left(\frac{\partial F_k}{\partial z_j} - \frac{\partial F_j}{\partial z_k}\right) + \frac{\partial}{\partial z_j}\left(\frac{\partial F_i}{\partial z_k} - \frac{\partial F_k}{\partial z_i}\right)$$

$$= \left(\frac{\partial^2 F_j}{\partial z_k \partial z_i} - \frac{\partial^2 F_i}{\partial z_k \partial z_j}\right) + \left(\frac{\partial^2 F_k}{\partial z_i \partial z_j} - \frac{\partial^2 F_j}{\partial z_i \partial z_k}\right) + \left(\frac{\partial^2 F_i}{\partial z_j \partial z_k} - \frac{\partial^2 F_k}{\partial z_j \partial z_i}\right) = 0$$

$\Rightarrow$ (2.5) 第一个方程成立}

总之：$*_1, *_2$成立 $\Rightarrow$ (2.5)成立 $\Leftrightarrow$ (2.4)是自伴的 （*Theorem* 2.2 内容）

注：$*_1, *_2$成立 $\Rightarrow$

$$K_{ij}\frac{dz_i}{dt} + D_i(z,t) = \left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right)\frac{dz_i}{dt} - \left(\frac{\partial B}{\partial z_i} + \frac{\partial F_i}{\partial t}\right) \quad (2.8)$$

（Birkhoff 系统）

证明：$*_1, \Omega = \sum_{i,j=1}^{2n} K_{ij} dz_i \wedge dz_j \Rightarrow d\Omega = 0$

Proof:

$$\Omega = \sum_{i,j}^{2n}\left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right) dz_i \wedge dz_j$$

$$= \sum_{i,j} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j - \sum_{i,j} \frac{\partial F_i}{\partial z_j}, dz_i \wedge dz_j$$

$$= \sum_{i,j} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j - \sum_{j,i} \frac{\partial F_j}{\partial z_i}, dz_j \wedge dz_i$$

$$= \sum_{i,j} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j + \sum_{j,i} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j$$

$$= 2 \sum_{i,j} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j$$

$$= 2 \sum_{i} \sum_{j} \frac{\partial F_j}{\partial z_i}, dz_i \wedge dz_j$$

$$= 2d\left(\sum_{k} F_k, dz_k \right) := 2d\alpha, \quad \alpha = \sum_{k} F_k, dz_k$$

$$\Rightarrow d\Omega = 2d^2\alpha = 0$$

数学前提：$d\Omega = 0$ 是 Birkhoff 系统存在的数学前提！

在哈密顿/Birkhoff 力学中，$d\Omega = 0$ 对应于泊松括号的 Jacobi 恒等式：

$$\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$$

保证哈密顿流（相空间的演化）保持辛形式不变（刘维尔定理），也就是相空间体积守恒；

保证泊松括号的代数结构是李代数，是量子力学对易关系的经典对应；

保证变分原理（哈密顿原理，Birkhoff 原理）的自洽性，不会出现矛盾的运动方程。

自伴性条件 $\Omega = \sum_{ij} K_{ij}\, dz_i \wedge dz_j$
$K_{ij} + K_{ji} = 0 \quad \leftrightarrow \quad \Omega$是反对称 $2-$形式
$\frac{\partial K_{ij}}{\partial z_k} + \frac{\partial K_{jk}}{\partial z_i} + \frac{\partial K_{ki}}{\partial z_j} = 0 \quad \Leftrightarrow \quad d\Omega = 0$ 的空间分量（Jacobi 恒等式） $\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$
$\frac{\partial K_{ij}}{\partial t} = \frac{\partial D_i}{\partial z_j} - \frac{\partial D_j}{\partial z_i} \quad \leftrightarrow \quad d\Omega = 0$ 的时间分量（时变系统的相容性）

$$\begin{cases} K_{ij} = \dfrac{\partial F_j}{\partial z_i} - \dfrac{\partial F_i}{\partial z_j} \\ D_i = -\dfrac{\partial B}{\partial z_i} - \dfrac{\partial F_i}{\partial t} \end{cases}$$

- 定理引用: Theorem (A), (B) ⇒ Page 23
- 系统优点: Birkhoff 系统的优点在于，其在表示非保守系统时仍能保持辛结构（并具有误差传播关系）。

Theorem(A)的证明：

(关键证明：$\frac{\partial \phi_w}{\partial t}=\frac{d}{dt}\hat{w}(\hat{z},z,t,t_0)-\frac{\partial \hat{w}}{\partial w}\frac{d}{dt}w(\hat{z},z,t,t_0)=\left(A_\alpha-\frac{\partial \hat{w}}{\partial w}\cdot C_\alpha\right)K^{-1}D(\hat{z},t)+\frac{\partial \alpha_1}{\partial t}-\frac{\partial \hat{w}}{\partial w}\frac{\partial \alpha_2}{\partial t}$)

已知：

$$\begin{cases}\hat{z}_i=g(t;z,t_0)\\ M=g_z=\dfrac{\partial \hat{z}}{\partial z}\\ \hat{w}=f(w,t,t_0)\\ N=f_N=\dfrac{\partial \hat{w}}{\partial w}\\ C_\alpha(z,z,t_0,t_0)+D_\alpha(z,z,t_0,t_0)\neq 0 \overset{\text{Theorem 3.5}}{\underset{\text{(3.2)成立}}{\Rightarrow}} C_\alpha(\hat{z},z,t,t_0)M+D_\alpha(\hat{z},z,t,t_0)\neq 0, |t-t_0| \text{足够小}\end{cases}$$

$\Rightarrow \exists$梯度映射$\hat{w}=f(w,t,t_0)$，满足$f_w=N$，$\exists!$生成函数$\phi(w,t,t_0)$，s.t.

$$f=\phi_N,\quad \alpha(\hat{z},z,t,t_0)=f(\alpha_2(\hat{z},z,t,t_0),t,t_0)=\phi_N(\alpha_2(\hat{z},z,t,t_0),t,t_0)$$
$$N=(A_\alpha M+B_\alpha)(C_\alpha M+D_\alpha)^{-1},$$
$$M=(A^\alpha N+B^\alpha)(C^\alpha N+D^\alpha)^{-1}.$$

$$\hat{z}=g(t;z,t_0),\quad (2.4)\text{ 重写成 } K(z,t)\frac{dz}{dt}=\nabla B+\frac{\partial F}{\partial t}\Rightarrow$$

$$\begin{cases}\dfrac{d\hat{z}}{dt}=K^{-1}(\hat{z},t)\left(\nabla B+\dfrac{\partial F}{\partial t}\right)(\hat{z},t)\\ \hat{z}|_{t=t_0}=z\end{cases}\quad \hat{w}=\alpha_1(\hat{z},z,t,t_0)\quad (P19\text{ 页下方})$$

其中$\hat{z}\longmapsto t$

$$\Rightarrow \frac{d\hat{w}}{dt}=\frac{\partial \hat{w}}{\partial \hat{z}}\cdot\frac{d\hat{z}}{dt}+\frac{\partial \hat{w}}{\partial z}\cdot\frac{dz}{dt}+\frac{\partial \hat{w}}{\partial t}$$
$$=A_\alpha K^{-1}\left(\nabla B+\frac{\partial F}{\partial t}\right)+B_\alpha\cdot 0+\frac{\partial \alpha_1}{\partial t}$$
$$=A_\alpha K^{-1}\left(\nabla B+\frac{\partial F}{\partial t}\right)+\frac{\partial \alpha_1}{\partial t}$$
$$\Rightarrow \frac{d\hat{w}}{dt}=\frac{\partial w}{\partial \hat{z}}\cdot\frac{d\hat{z}}{dt}+\frac{\partial w}{\partial t}=C_\alpha K^{-1}\left(\nabla B+\frac{\partial F}{\partial t}\right)+\frac{\partial \alpha_2}{\partial t},\quad \hat{w}=f(w,t,t_0)$$

其中$w\longmapsto t$

$$\frac{d\hat{w}}{dt}=\frac{\partial \hat{w}}{\partial w}\frac{dw}{dt}+\frac{\partial \hat{w}}{\partial t}\Rightarrow$$

$$\frac{\partial \hat{w}}{\partial t} = -\frac{\partial \hat{w}}{\partial w}\frac{dw}{dt} + \frac{d\hat{w}}{dt} = \frac{d\hat{w}}{dt} - \frac{\partial \hat{w}}{\partial w}\frac{dw}{dt}$$

$$= A_\alpha K^{-1}\left(\nabla B + \frac{\partial F}{\partial t}\right) + \frac{\partial \alpha_1}{\partial t} - \frac{\partial \hat{w}}{\partial w}\left[C_\alpha K^{-1}\left(\nabla B + \frac{\partial F}{\partial t}\right) + \frac{\partial \alpha_2}{\partial t}\right]$$

$$= \left(A_\alpha - \frac{\partial \hat{w}}{\partial w}C_\alpha\right)K^{-1}\left(\nabla B + \frac{\partial F}{\partial t}\right) + \frac{\partial \alpha_1}{\partial t} - \frac{\partial \hat{w}}{\partial w}\frac{\partial \alpha_2}{\partial t} \quad (4.5)$$

$$\frac{\partial \hat{w}}{\partial w} = \frac{\partial \hat{z}}{\partial w} = f_w = N \text{ 可逆},$$

无法写成 w $= ?(\hat{w}) \Rightarrow$

$\frac{\partial \hat{w}}{\partial t} := \bar{A}\left(\bar{z}, z, \frac{\partial \hat{w}}{\partial w}, t, t_0\right) := \mathcal{A}\left(\hat{w}, w, \frac{\partial \hat{w}}{\partial w}, t, t_0\right) = \mathcal{A}(\phi_w, w, \phi_{ww}, t, t_0)$.（证毕）

Theorem (B) 证明:

$$\phi_w = \hat{w} = f(w, t, t_0), \quad \phi_w^{(0)} = f(w, t_0, t_0)$$

$$\frac{\partial \phi_w}{\partial t} = \mathcal{A}(\phi_w, w, \phi_{ww}, t, t_0)$$

$$= \mathcal{A}(\phi_w^{(0)}, w, \phi_{ww}^{(0)}, t_0, t_0) + \sum_{k=1}^{\infty}\frac{1}{k!}(t-t_0)^k D_t^k \mathcal{A}(\phi_w^{(0)}, w, \phi_{ww}^{(0)}, t_0, t_0),$$

$$\phi_w = \phi_w^{(0)}(w, t_0) + \sum_{k=P}^{\infty}(t-t_0)^{k+1}\phi_w^{(k+1)}(w, t_0) \Rightarrow \frac{\partial \phi_w}{\partial t} = \sum_{k=0}^{\infty}(k+1)(t-t_0)^k \phi_w^{(k+1)}(w, t_0)$$

$$\Rightarrow \begin{cases} \phi_w^{(0)}(w, t_0) = f(w, t_0, t_0) \\ \phi_w^{(k+1)}(w, t_0) = \dfrac{1}{(k+1)!} D_t^k \mathcal{A}(\phi_w^{(0)}, w, \phi_{ww}^{(0)}, t_0, t_0) \end{cases}$$

底部总结（绿色圈标注）

$Theorem\ (A), (B) \Rightarrow Page\ 23$ (36)

好处保持了用 Birkhoff 系统表示的非保守系统的辛结构（与误差相联系）

说明此方法的优势，对减小误差的作用。